



ADI *Power*

Power Management Fundamentals II Session 1: Inside LDO and Buck Regulators

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analog.com

Agenda

- 1 Overview
- 2 Linear regulators (LDOs)
- 3 Buck Regulators
- 4 Differentiating Technologies

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- 1 Overview
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Powering Up – Module Overview

Session 1

LDOs & Bucks

LDO
Regulators

Buck
Regulators

Takeaway

A Deeper Look
into the workings
of LDOs & Bucks

Session 2

Boost &
Buck-Boost

Boost
Regulators

Buck-Boost
Regulators

Takeaway

A Deeper Look into
the workings of
Boost & Buck-Boost

Session 3

Sepic, Ćuk &
Inverters

Sepic/Ćuk

Inverting
Regulators

Takeaway

A Deeper Look into
the workings of
Sepic/Ćuk &
Inverters

Session 4

Isolated
Converters

Flyback
Converters

Forward
Converters

Takeaway

A Deeper Look at
Isolated Converters

Session 5

Power Path, Hot-swap
& Surge-Stopper

Power Path

Hot-Swap

Surge-
Stopper

PoE

Takeaway

A Deeper Look into
Power Input
Circuits

Session 6

Layout
Considerations

Layout

Takeaway

A Deeper Look into
Layout
Considerations

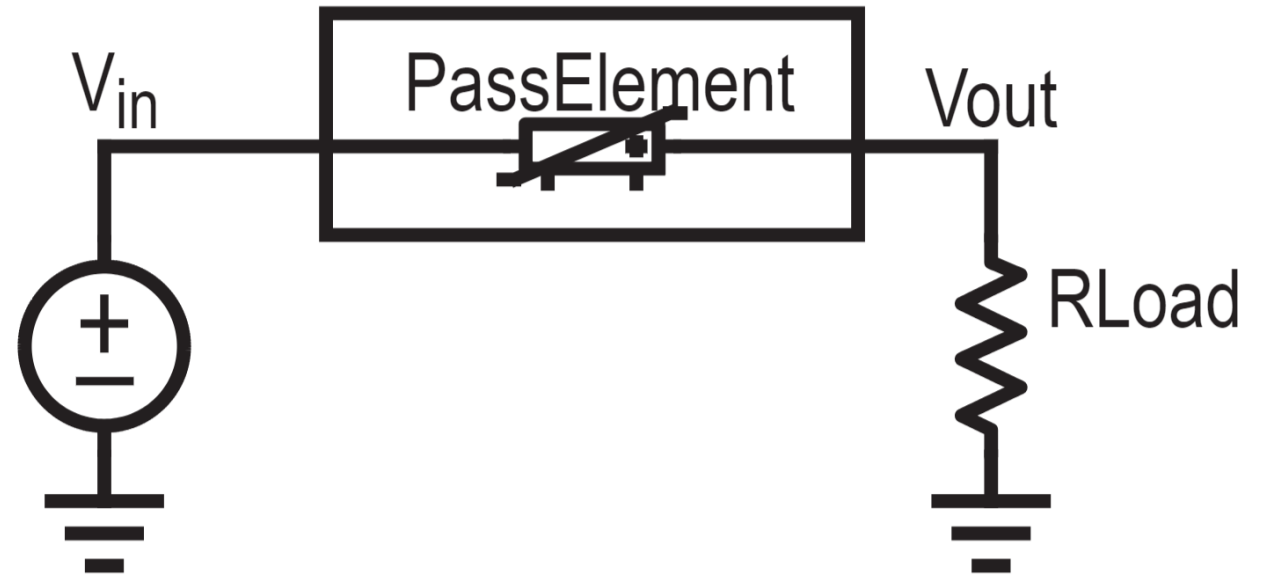
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LDO: Operation

Theory of operation.

- Changing the Pass Element resistance changes V_{out} .
- Pass Element Dispenses excess energy as heat.



$$V_{in} \times \frac{R_{load}}{R_{load} + R_{passElement}} = V_{out}$$

LDO: Pass Transistor

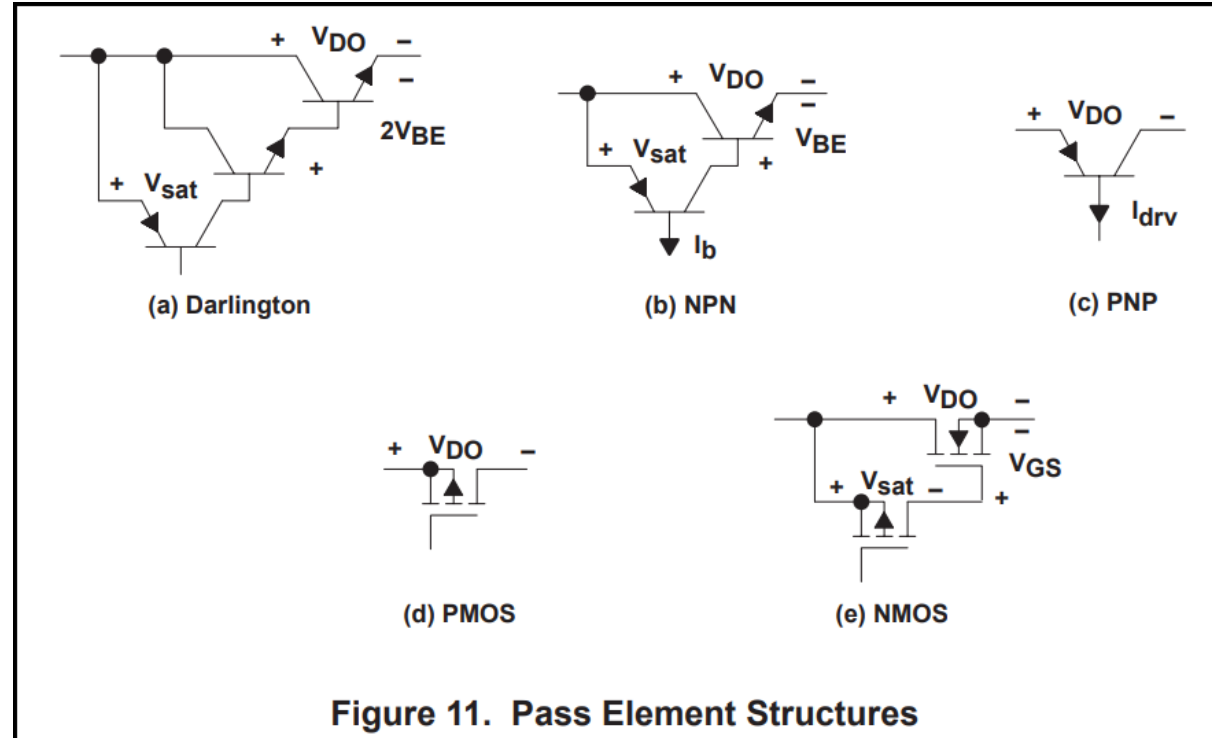
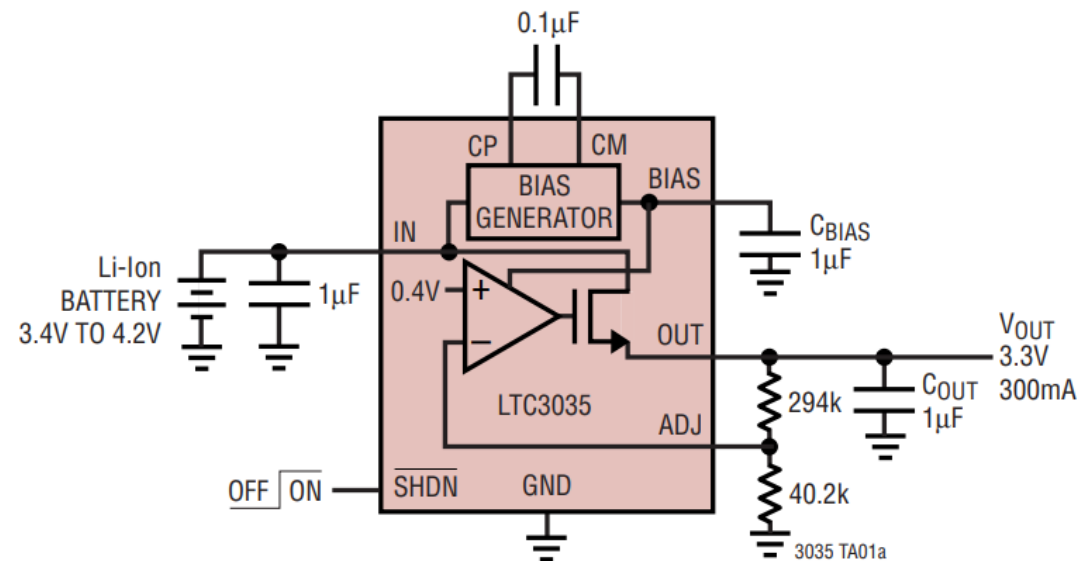


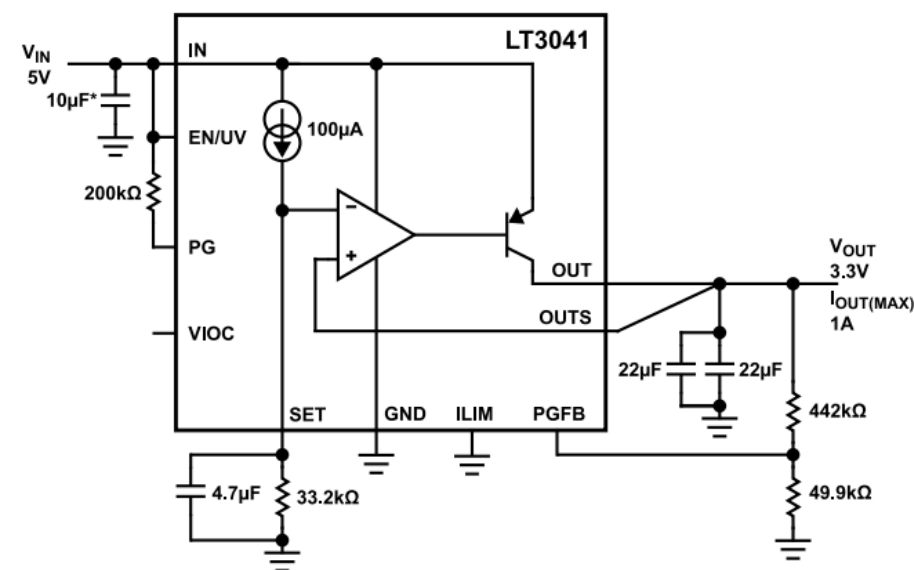
Table 1. Comparison of Pass Element Structures

PARAMETER	DARLINGTON	NPN	PNP	NMOS	PMOS
$I_{o,max}$	High	High	High	Medium	Medium
I_q	Medium	Medium	Large	Low	Low
$V_{dropout}$	$V_{sat} + 2V_{be}$	$V_{sat} + V_{be}$	$V_{ce(sat)}$	$V_{sat} + V_{gs}$	$V_{SD(sat)}$
Speed	Fast	Fast	Slow	Medium	Medium

LDO: Control Methods



Voltage Mode Control



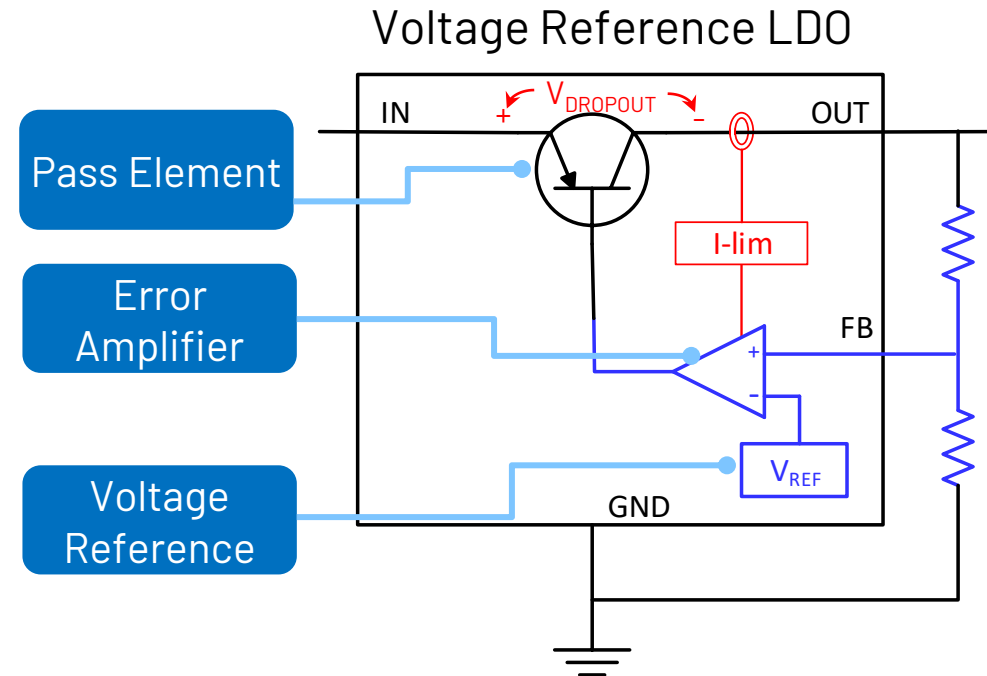
*OPTIONAL, SEE APPLICATIONS INFORMATION.

Current Mode Control

LDO: Voltage Mode Control

Voltage Reference LDOs

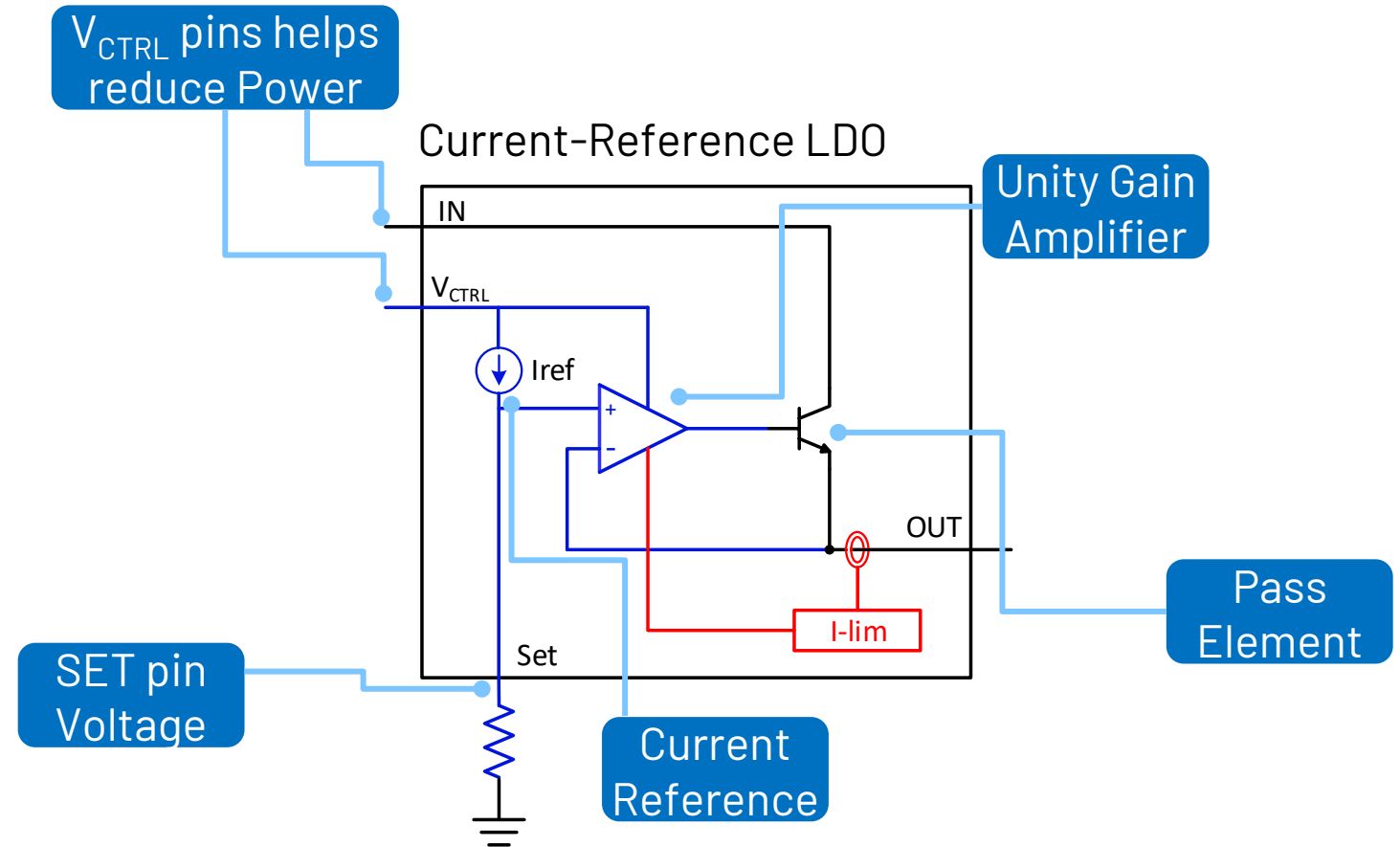
- Pass Element
 - BJT or MOSFET
- Control Circuit
 - Error Amplifier – Gain set by FB resistor divider
 - Current Limit
- Reference
 - Typically 0.5V to 1V
- Setting the output voltage
 - FB Pin
- Limitations
 - $V_{\max} = V_{\text{IN}} - V_{\text{DROPOUT}}$
 - $V_{\min} = V_{\text{REF}}$



LDO: Current Mode Control

Current Reference LDOs

- Pass Element
- Control Circuit
 - Error Amplifier – Unity Gain Buffer
 - Current Limit
- Reference
 - Constant Current (10-100 μ A)
 - Develops voltage at SET pin
- Limitations
 - $V_{max} = V_{in} - V_{dropout}$
 - $V_{min} \cong 0V$
- Separate IN and V_{CTRL} pins

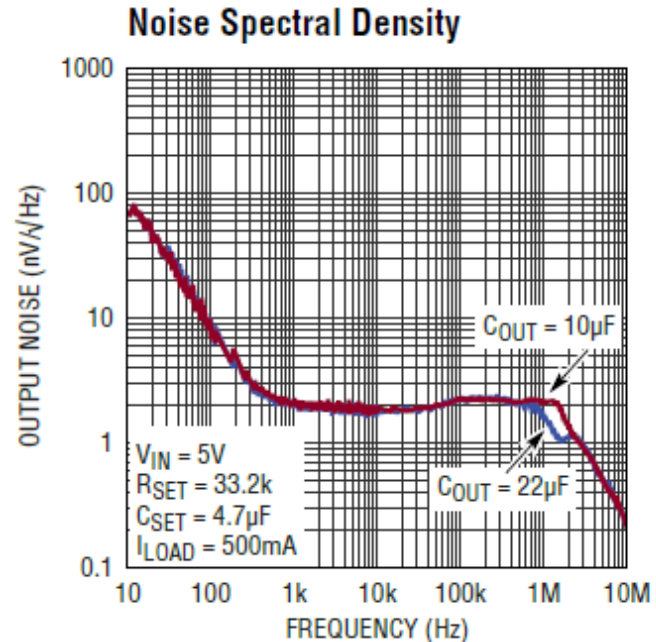


$V_{CONTROL}$ Dropout Voltage (Note 4)	●	1.2	1.6	V
		1.35		V
V_{IN} Dropout Voltage (Note 4)	●	100	200	mV
		350	500	mV

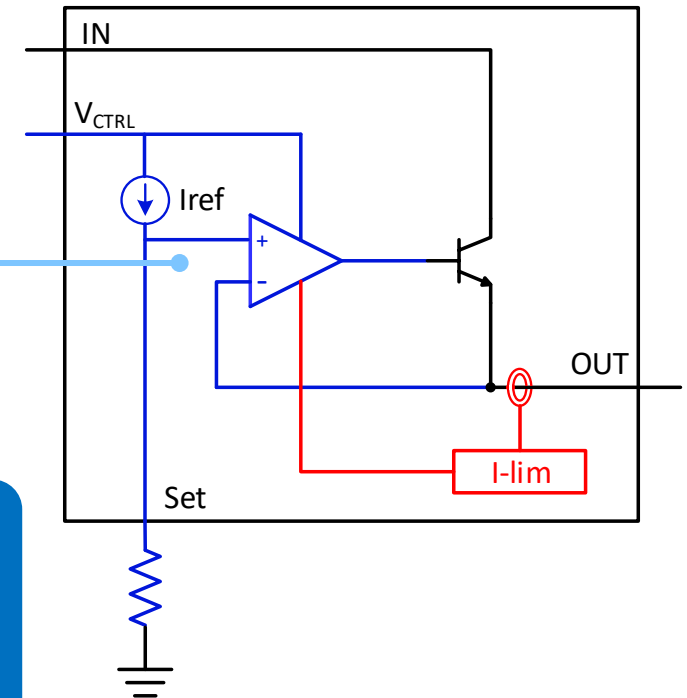
LDO: Current Mode Control

Current Reference LDOs

- Can be very low noise – 0.8 μ Vrms
- Because the Error Amplifier is unity gain, the noise is independent of V_{OUT}
- Many of these Current Reference LDOs are very low noise parts – ideal for noise-sensitive applications



Current Reference LDO



Error Amp is a unity-gain buffer. Noise independent of V_{OUT}

LDO: Current Mode Control

Current Reference LDOs

- V_{OUT} tolerance only due to V_{OS} in Error Amp, and Independent of V_{OUT}
- Parallellable with very small ballast R – only needs to compensate for V_{OS}
- Paralleling helps with thermals – heat spread across multiple IC packages

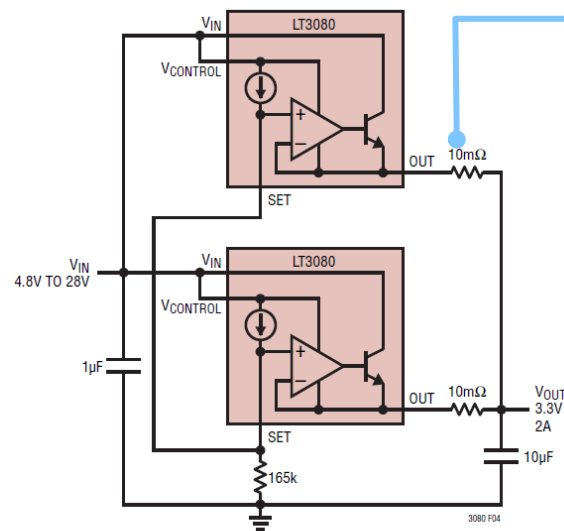
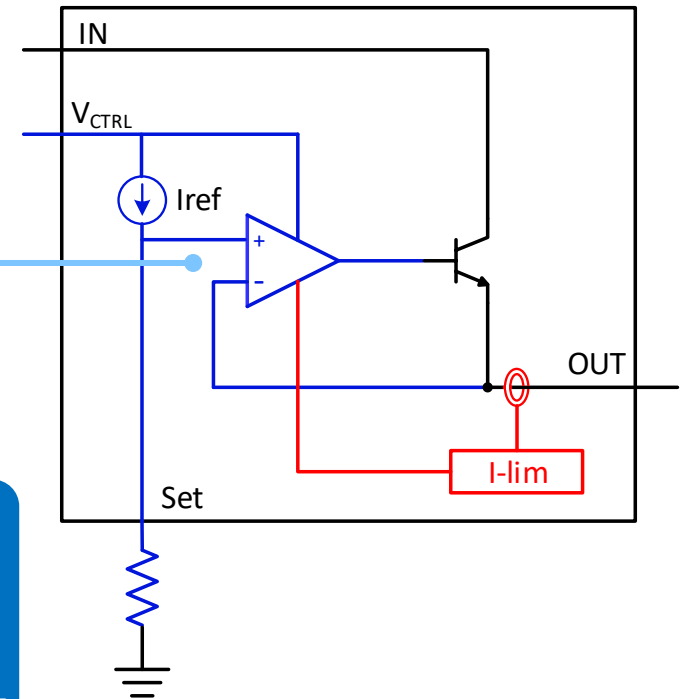


Figure 4. Parallel Devices

Only 10mΩ of ballast needed

Error Amp
Offset Voltage
 $\pm 3.5\text{mV}$ over Temp
Independent of V_{OUT}

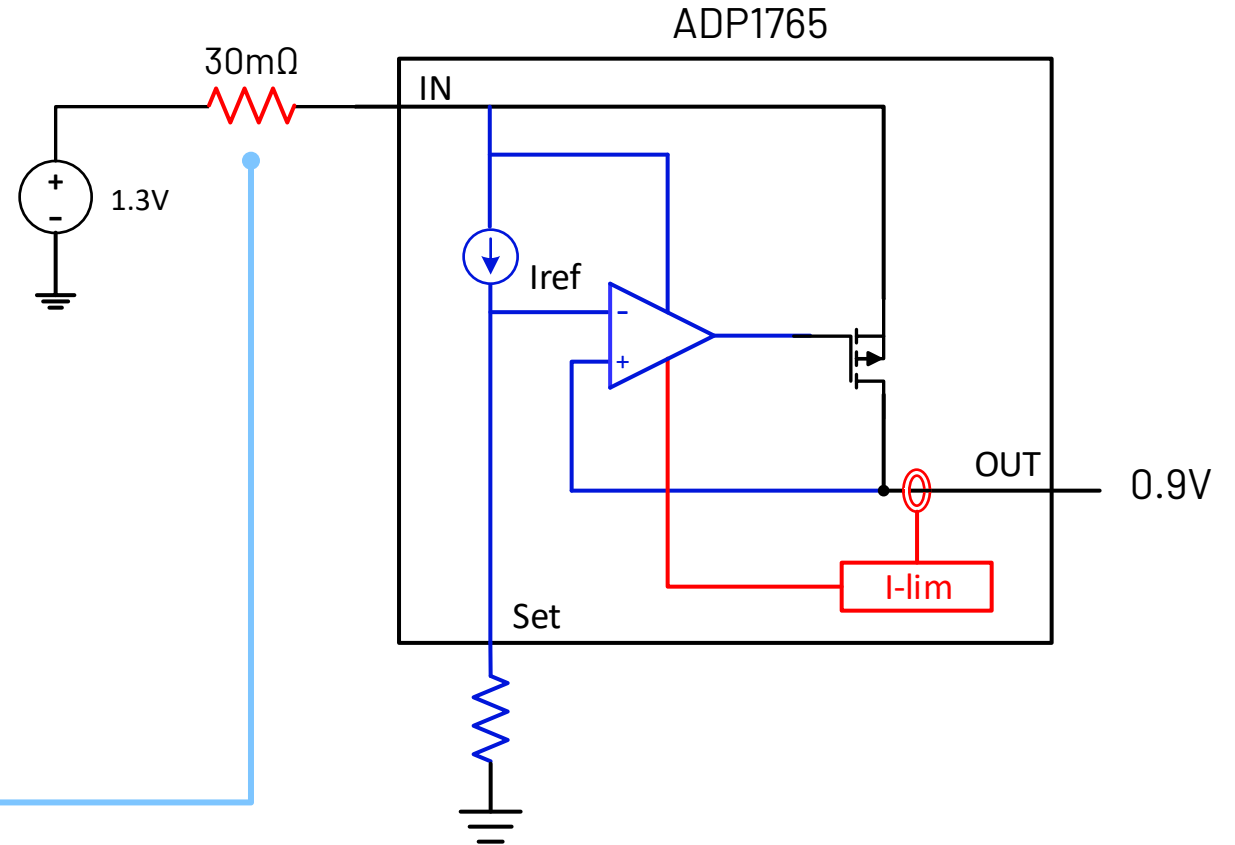
Current-Reference LDO



LDO: Thermal Management

High Current Example

- ADP1765 – 5A Current Reference LDO rated for 125°C operation
 $V_{\text{DROPOUT}} = 95\text{mV}$ (w/c over temp)
- $V_{\text{IN}} = 1.3\text{V}$, $V_{\text{out}} = 0.9\text{V}$
- $T_{\text{ambient}}(\text{max}) = 60^\circ\text{C}$
- $T_j = T_{\text{Amb}} + PD * \theta_{JA}$
- $PD = (V_{\text{IN}} - V_{\text{OUT}}) * I_{\text{out}}$, $\theta_{JA} = 40^\circ\text{C/W}$
- $T_j = 60^\circ\text{C} + (1.3 - 0.9) * 5 * 40 = 140^\circ\text{C}$
- This exceeds T_{JMAX} for the ADP1765!
- What to do?
- Add a resistor in series: 30mΩ, 1.5W (150mV drop @5A)
 - Reduces IN voltage to 1.15V, so
 - $T_j = 60^\circ\text{C} + (1.15 - 0.9) * 5 * 40 = 110^\circ\text{C}$



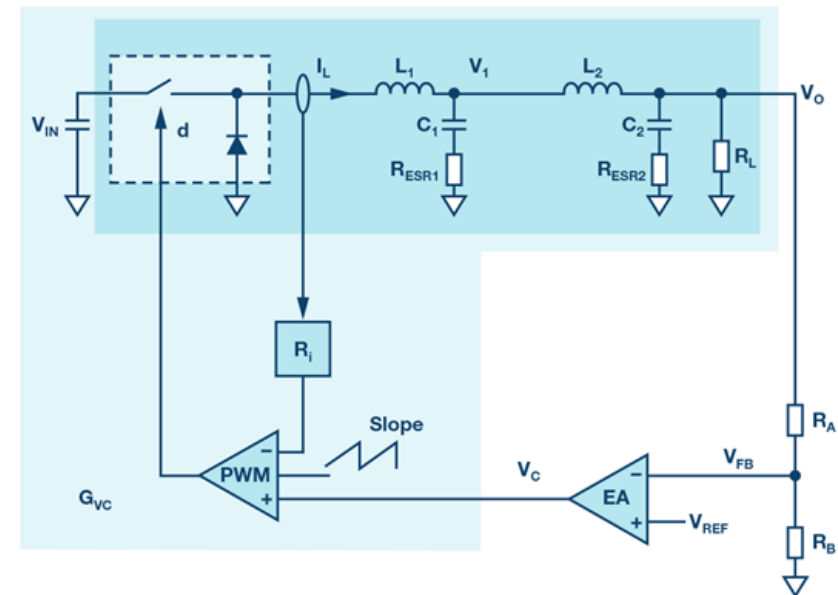
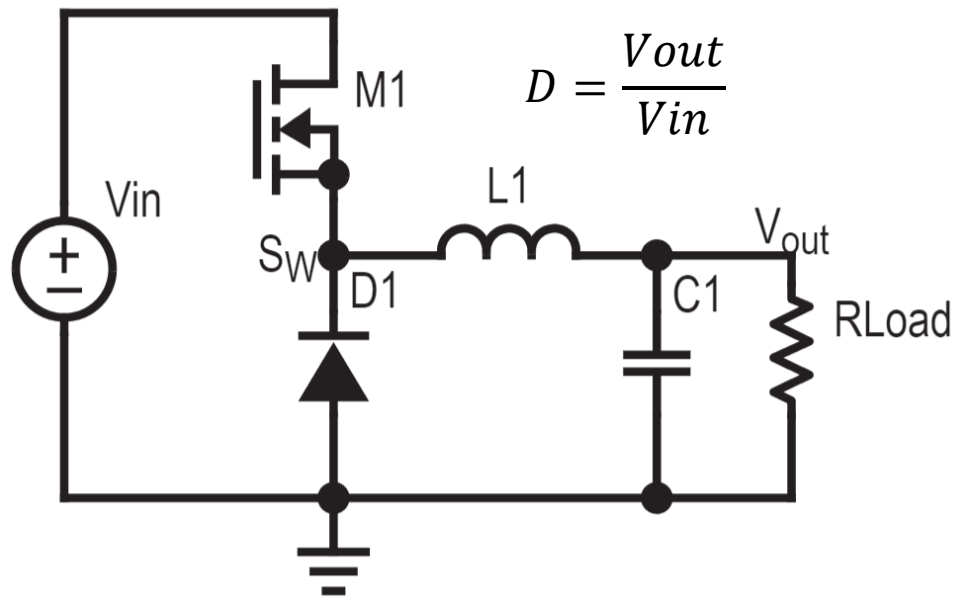
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- 3 Buck Regulators**
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Buck Regulators: Operation

Theory of operation.

- Uses a power switch, an inductor, and a diode to transfer energy from input to output
- Varies the time turning On/Off the switch to achieve the target V_{out} voltage

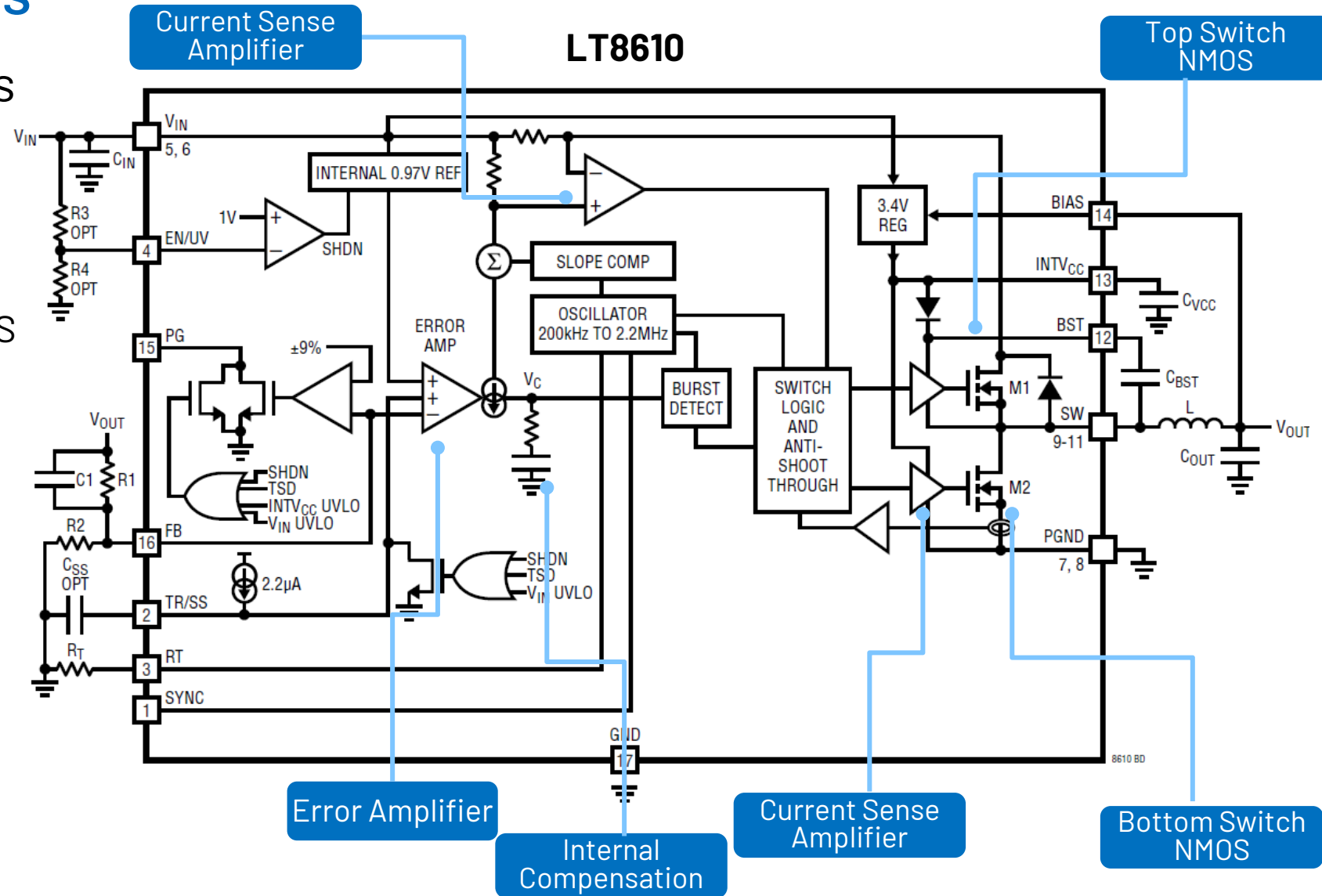


Current Mode Control

Buck Regulators: Parts

Buck (Step-down) Switchers

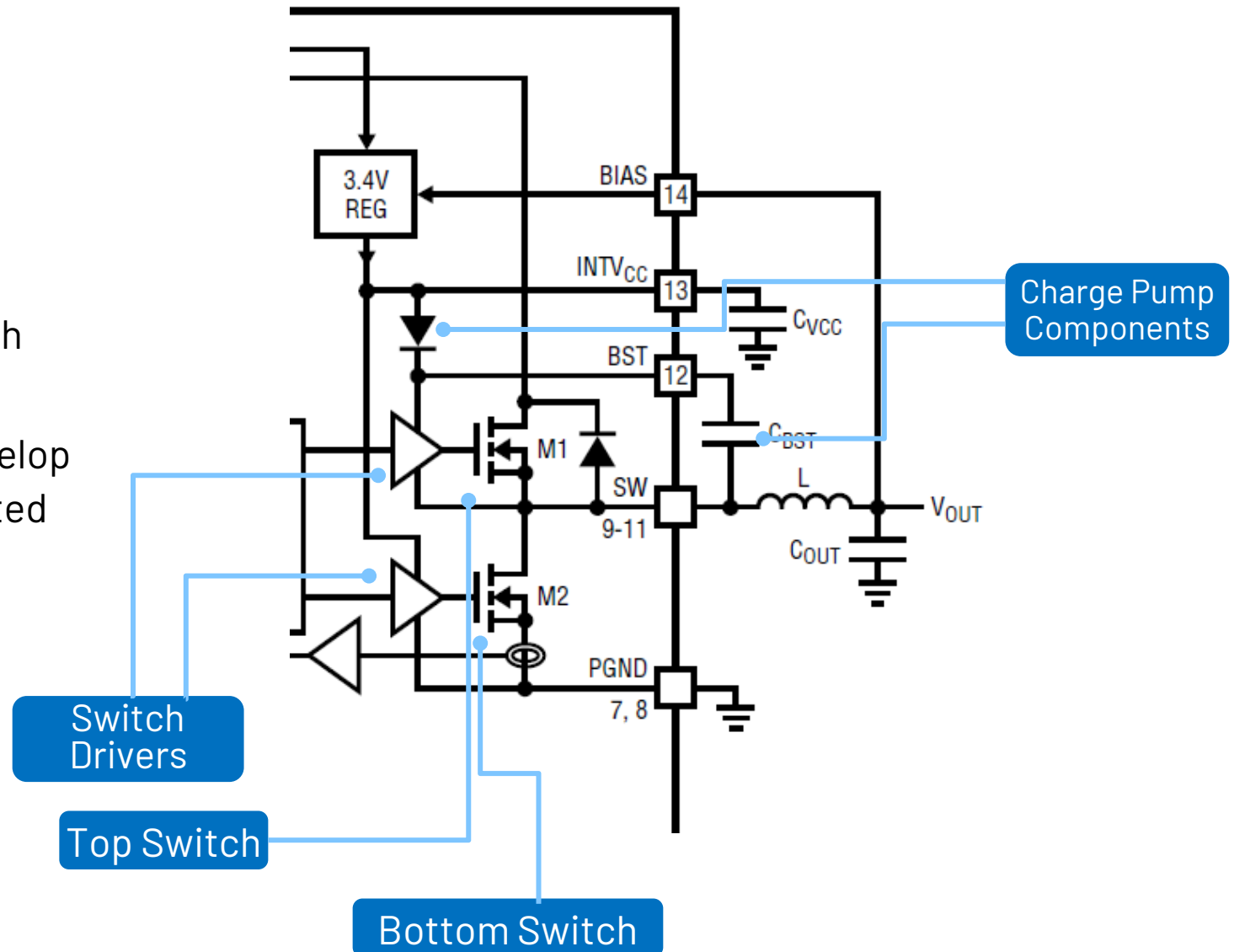
- Switches
 - Top Switch: NPN, PNP, PMOS, NMOS
 - Bottom Switch: Diode, NMOS
- Example: LT8610



Buck Regulators: Switch Drives

Buck Switch Drive

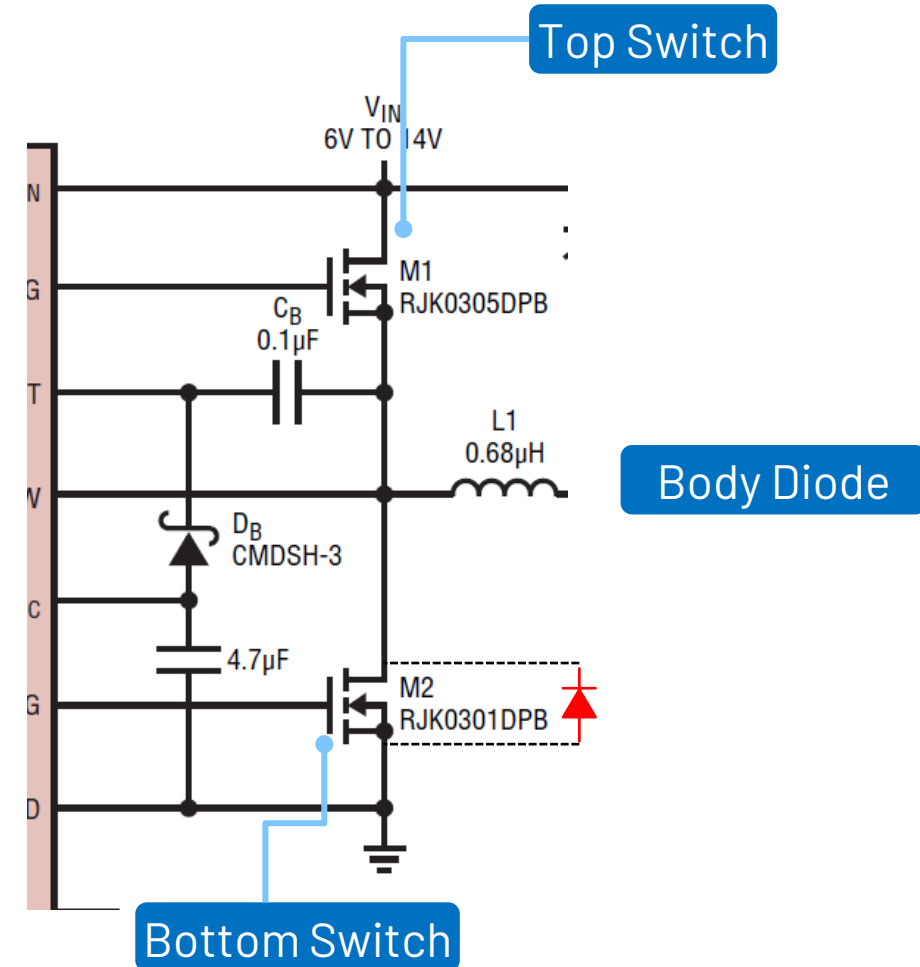
- Bottom Switch is GND-referenced Driver powered from $INTV_{CC}$ to GND
- Top Switch requires Boosted Drive Voltage must be above V_{IN} if Top Switch is NPN or NMOS
- Boost pin forms a Charge Pump to develop higher voltage: Diode and Cap connected between $INTV_{CC}$ and SW
Boost generates voltage = $V_{IN} + INTV_{CC}$
- Notes:
Others might refer to the Boost pin as "Boot"
If the Top Switch is PNP or PMOS, then no Boost is required



Buck Regulators: Switch

Buck Components

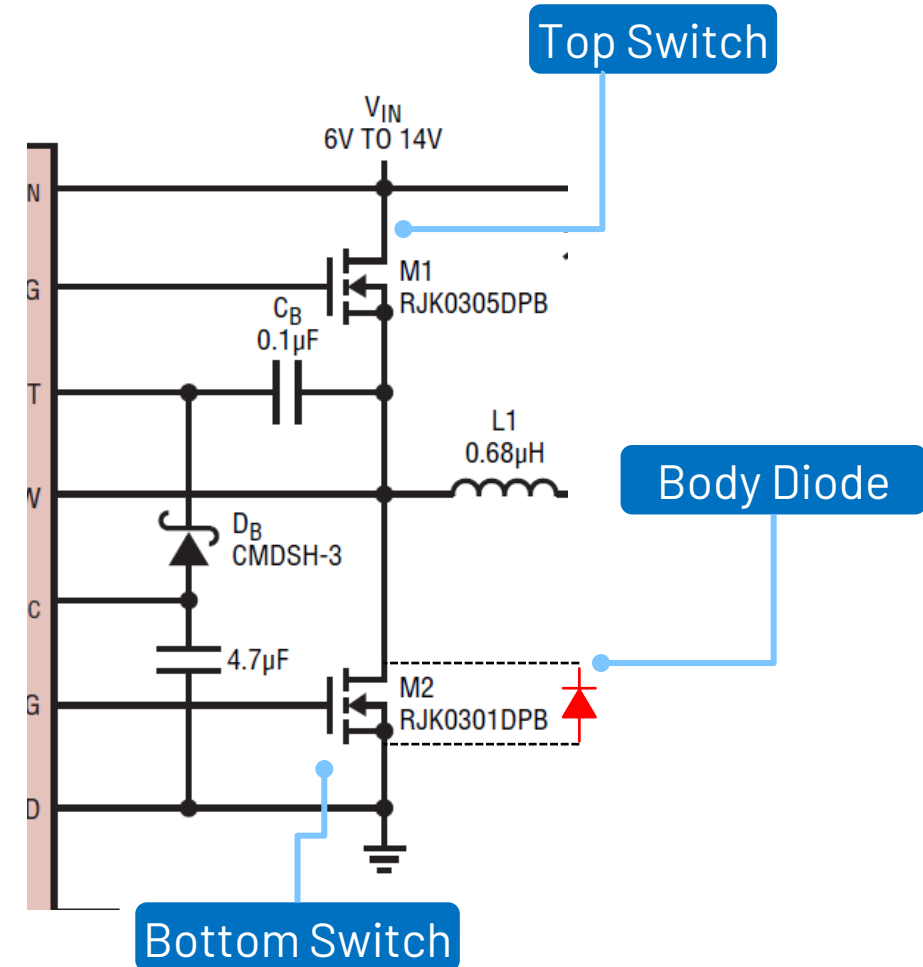
- FETs
 - Top & Bottom FETs are different parts Why?
 - Top FET must pull SW node up from GND – total V_{DS} swing is V_{in} . During this time, there is power loss in FET and Driver
 - When the Top FET is turned off, the SW node naturally commutates down below GND due to the inductor current, and Body Diode conducts. This is called the Dead Time.
 - Driver turns Bottom FET on once SW is below GND. FET shorts out Body Diode, reducing conductive loss. Total V_{DS} swing is only $\sim 0.7V$, in much shorter time. Switching loss is extremely small as a result



Buck Regulators: Switch

Buck Components

- FETs
 - Top FET is selected for low Q_g or a compromise between Q_g (switching loss) and R_{dsON} (conductive loss). It depends on the V_{IN}/V_{OUT} ratio.
 - Bottom FET has very little switching loss, so choose for low R_{dsON}
 - Example in LTPowerCAD: LTC3855 1.2Vout



Buck Regulators: Compensation

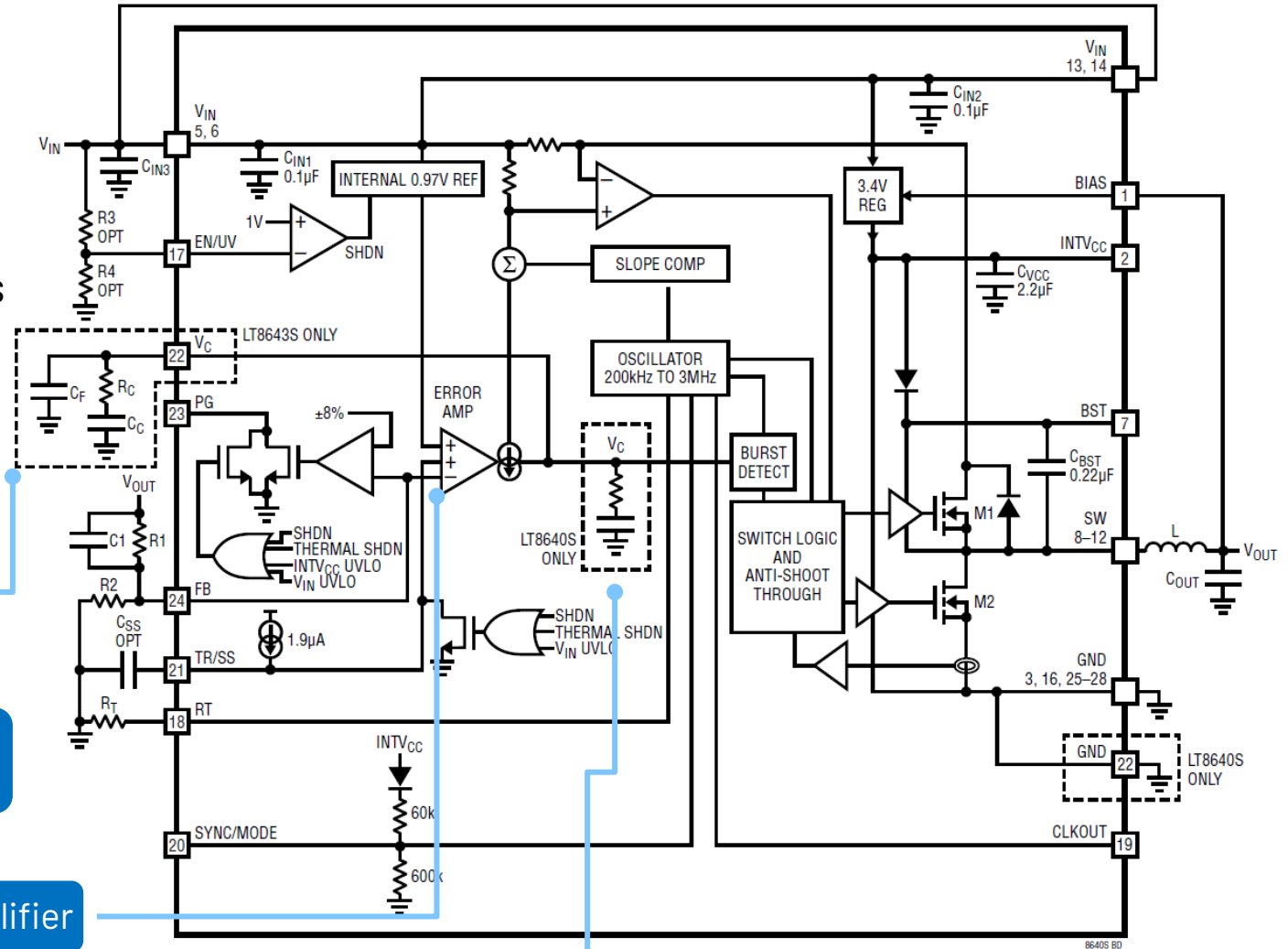
Buck feedback control

- Many Monolithic Bucks have internal compensation
- Necessary for very low bias current devices
Internal comp can use very large R values (possible on-chip, problematic on PCB)
- A few offer external compensation
- Trade-off is higher bias current
LT8640S: 1.7uA (Internal Comp)
LT8643S: 230uA (External Comp)

External
Compensation
LT8643S

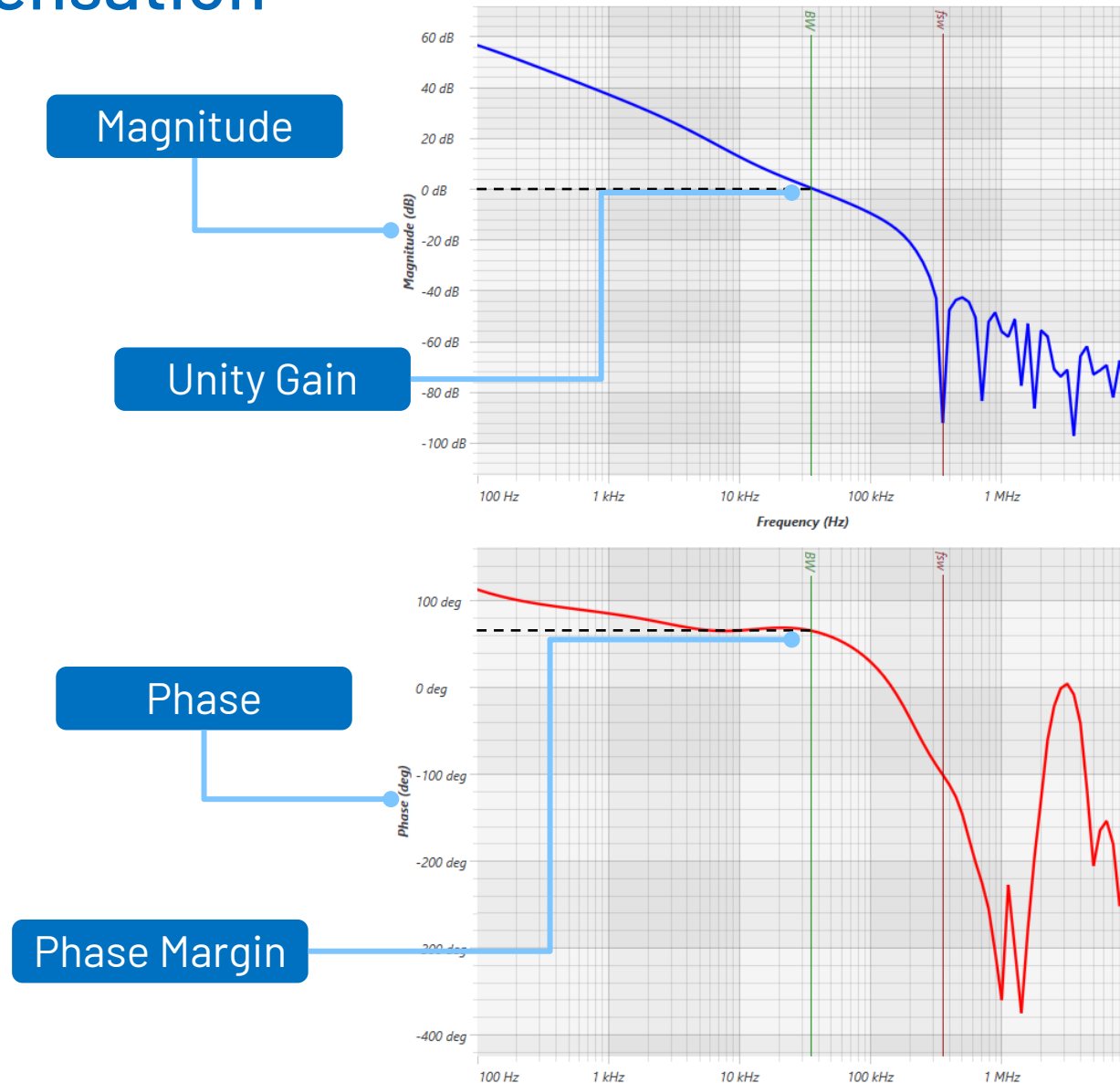
Error Amplifier

Internal
Compensation
LT8640S

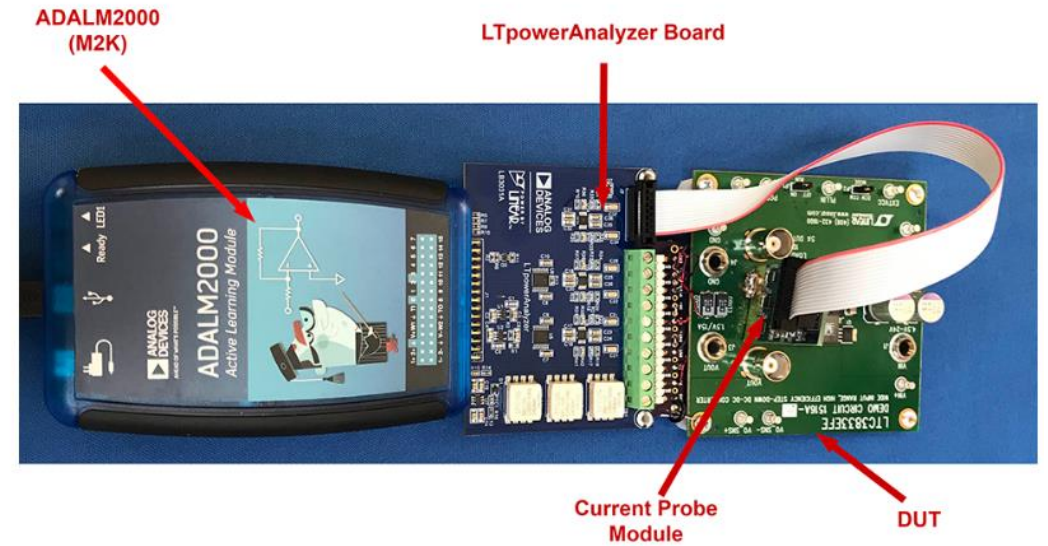
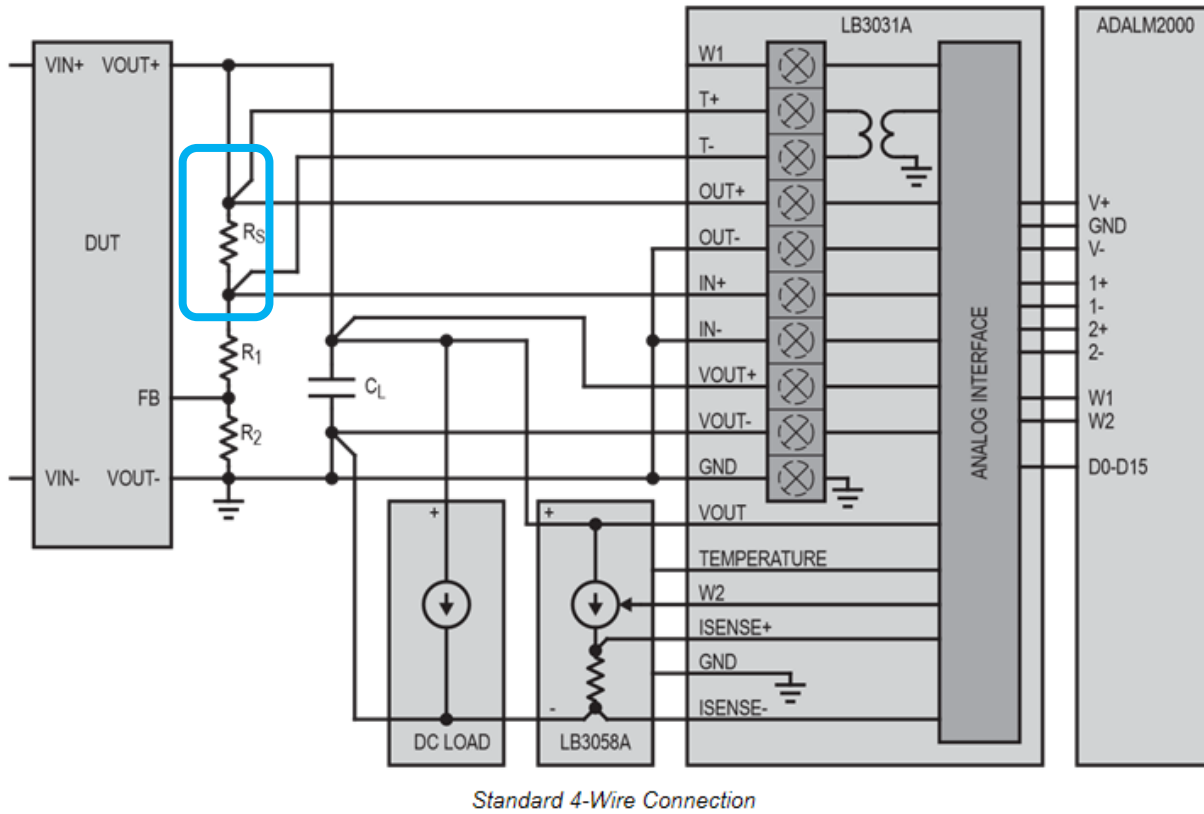


Buck Regulators: Compensation

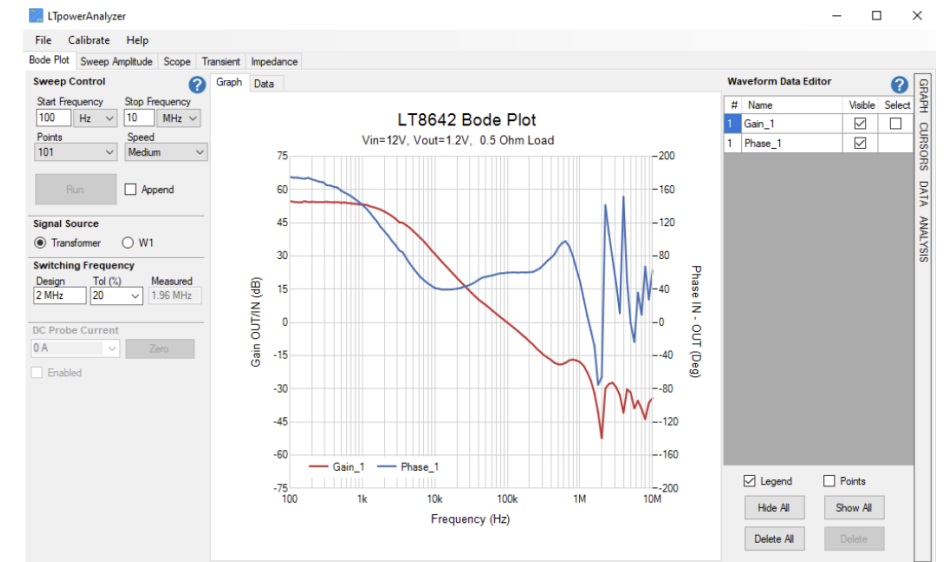
- Compensation – refer to Bode Plot
- Magnitude Plot – Unity Gain point is the frequency where the gain drops to 0dB
- Phase Plot – Phase at 0dB frequency is the Phase Margin
- Per Nyquist, if F_{SW} is 350kHz, then Loop Bandwidth must be less than 175kHz. In practice, we typically use ~10-20% of F_{SW} or 35-70kHz for this example
- Adjust compensation to get appropriate bandwidth and good phase margin (ideally 60° or more)



Buck Regulators: Compensation



LTpowerAnalyzer System and ADALM2000 (M2K) Module

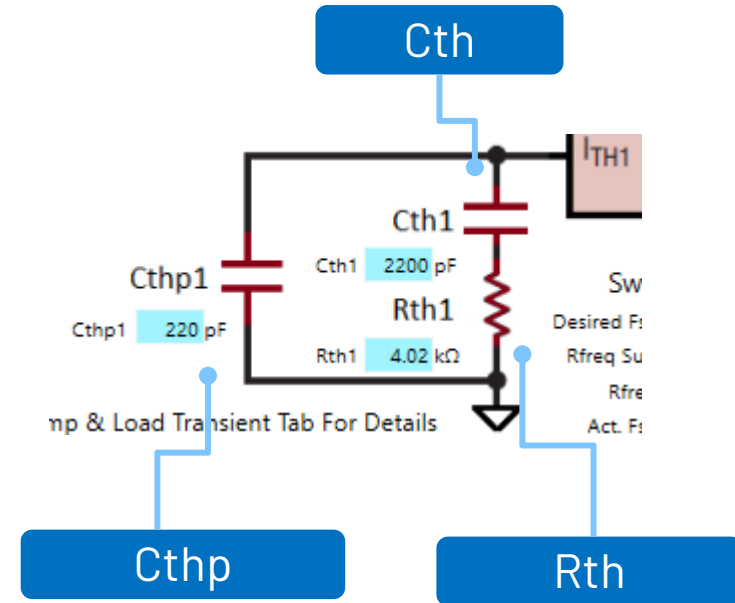


Buck Regulators: Compensation

Buck Compensation Components

- Current mode controllers use Type-II compensation
Cth, Rth and Cthp components
- Cth affects low-frequency phase response
- Rth affects gain (raises or lowers entire Gain curve), which can affect 0dB gain frequency
- Cthp affects high-frequency phase response

- Important Note:
A switching regulator is a sampled-data system, just like an ADC
- Nyquist requirement: the bandwidth of the regulator loop must be lower than $\frac{1}{2} F_{SW}$, or aliasing will occur
- This is critical when choosing compensation components



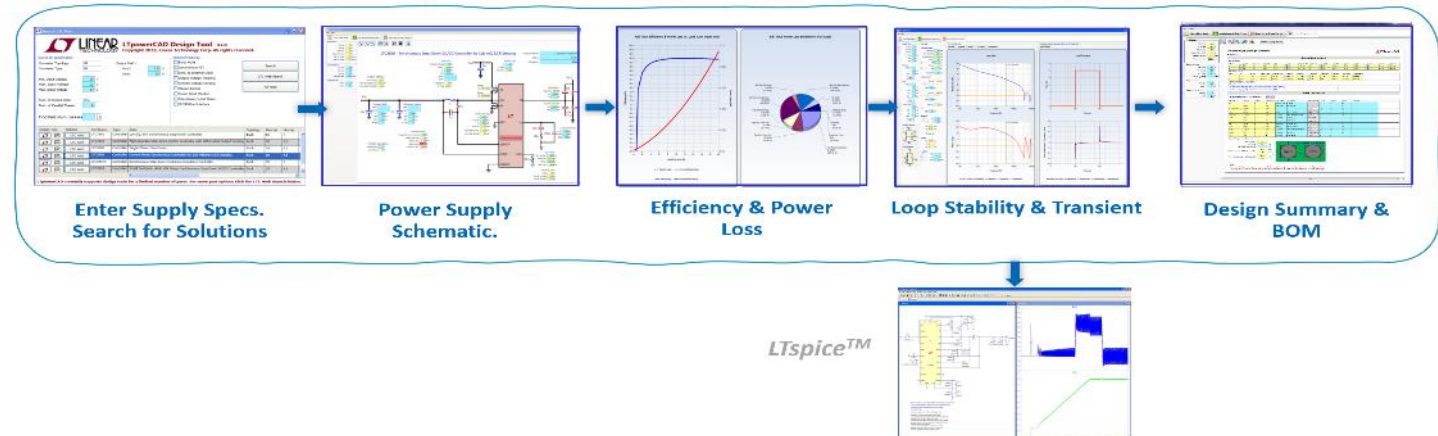
Buck Regulators: Compensation



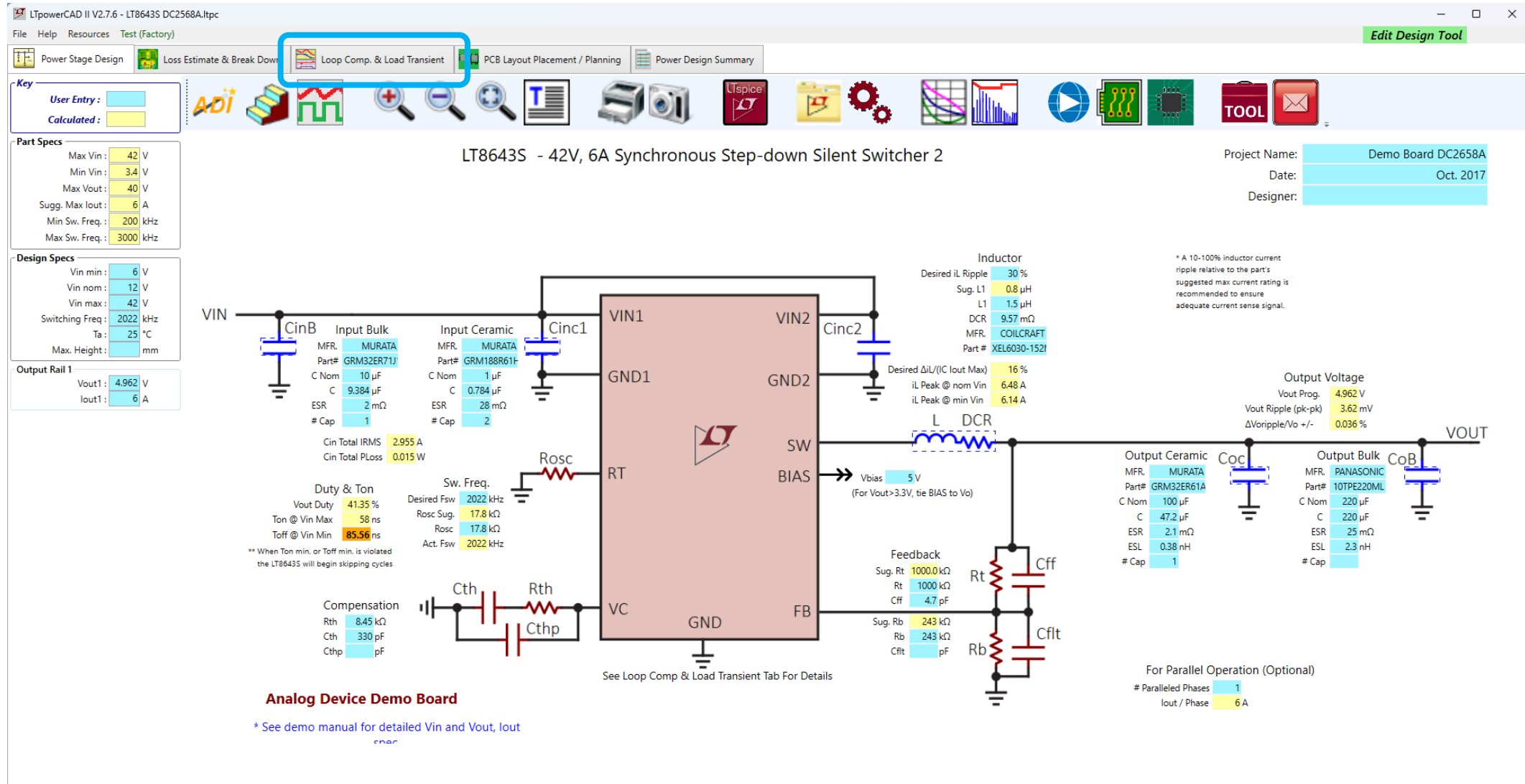
A Complete, Step-by-Step Power Converter Design Tool



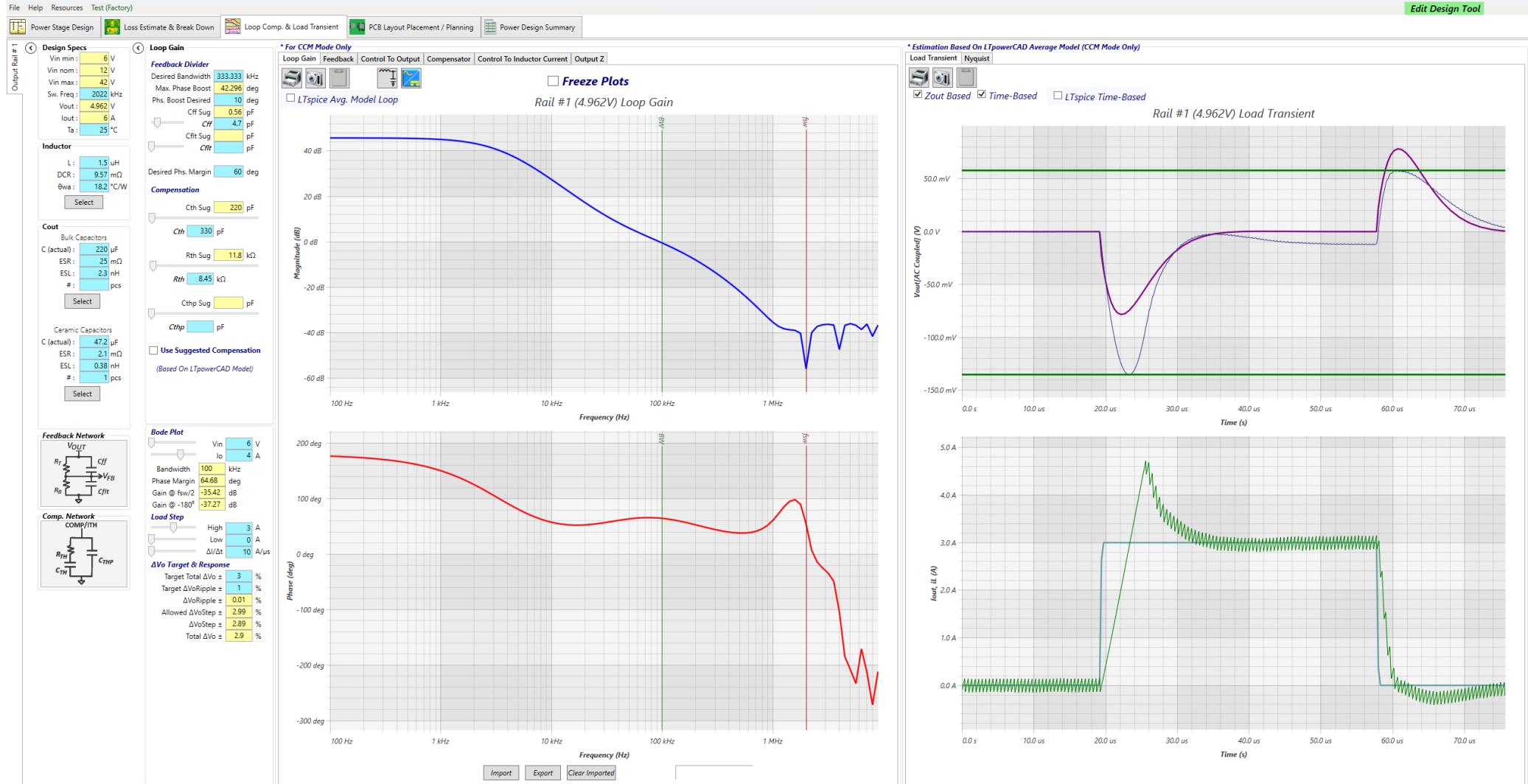
Design a Power Converter In Five Simple Steps



Buck Regulators: Compensation



Buck Regulators: Compensation

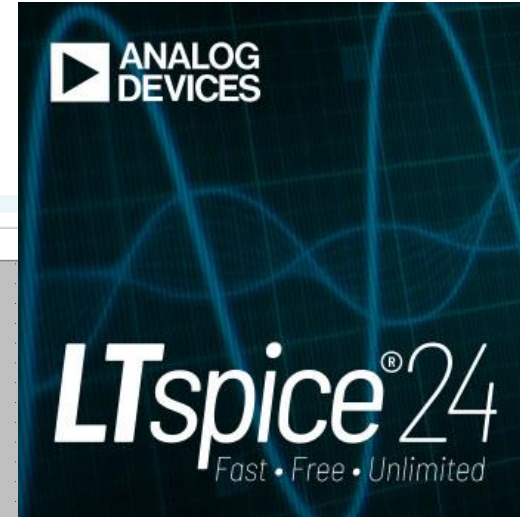
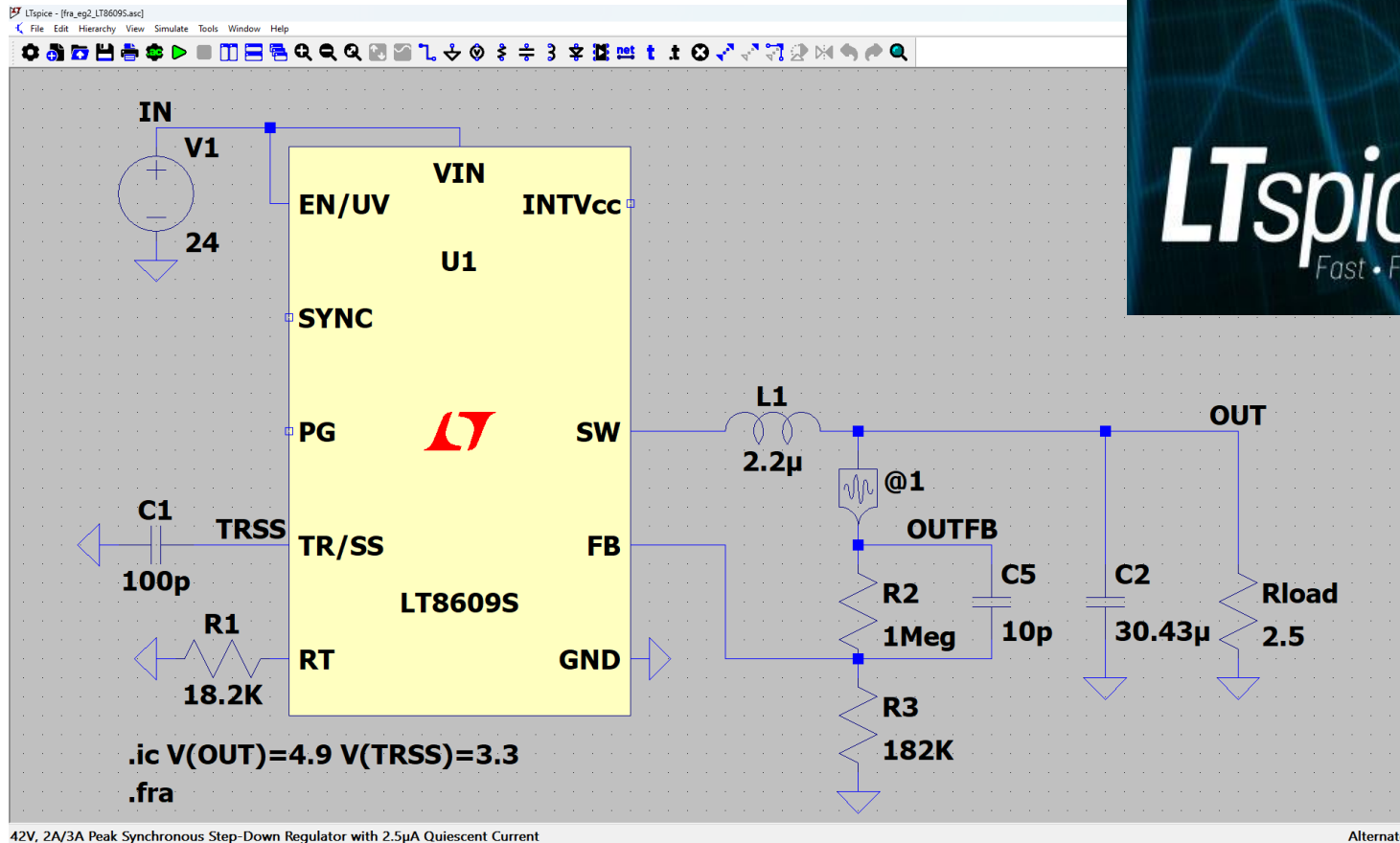


Buck Regulators: Compensation

Important Note:

- Choice of output capacitors affects loop response
ESR in caps creates a zero in the response
This adds phase boost to the loop. ESR zero is at $\frac{1}{2\pi RC}$
- Ceramic cap: 100 μ F, ESR = 2m Ω - Zero at 800kHz - this is above practical loop bandwidth
Polymer cap: 470 μ F, ESR = 9m Ω - Zero at 37kHz - this is well within the normal loop bandwidth
- Example in LTPowerCAD - LTC3855 1.2V

Buck Regulators: Compensation



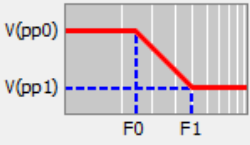
Buck Regulators: Compensation

Frequency Response Analyzer - @1

Analysis Type
☒ Gain vs. Frequency ☐ Impedance vs. Frequency

Help Me Configure This for a Switching Regulator

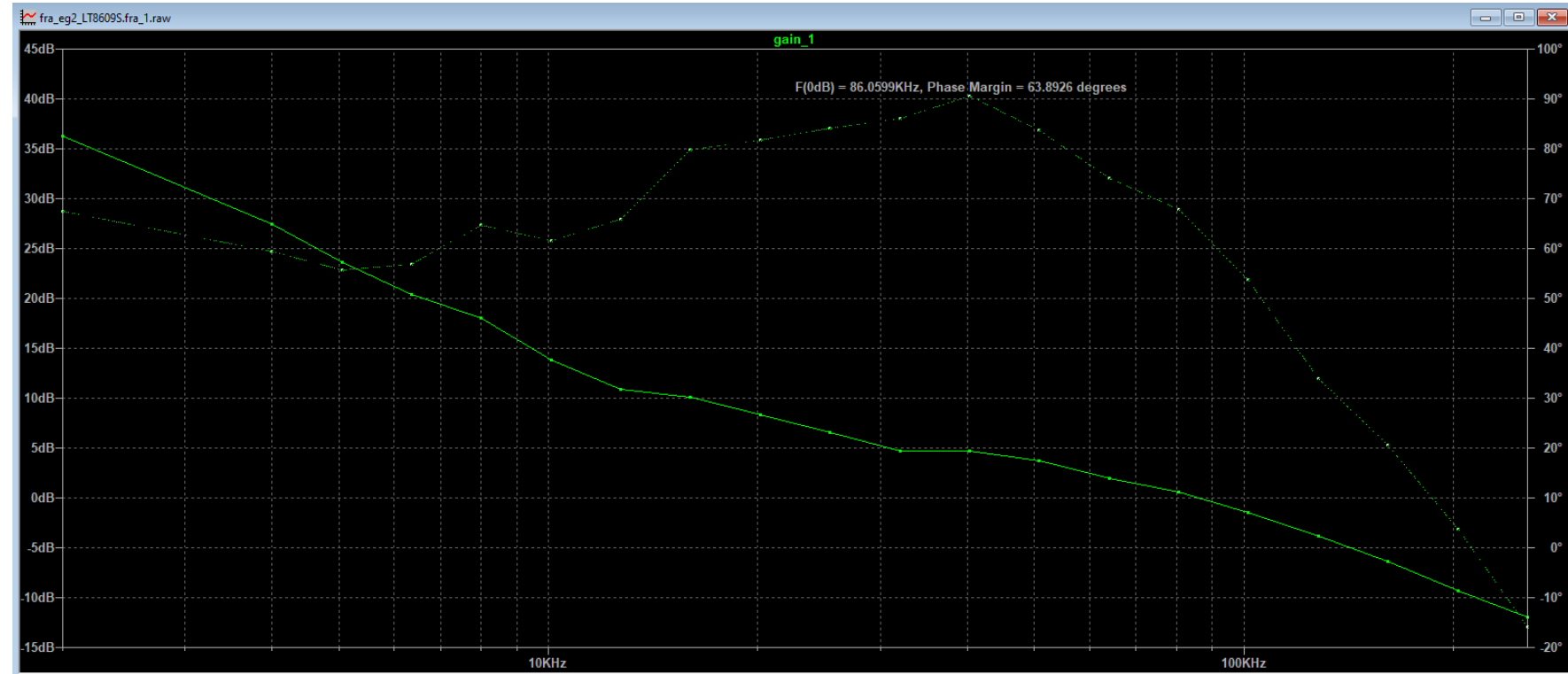
Stimulus Frequency
 Start at Frequency[Hz]*: 2k
 End at Frequency[Hz]*: 256k
 3 points per octave (10 points per decade)
 Coarse Steps up to Frequency[Hz]: 4k
 Add These Specific Frequencies[Hz]:
 Max # of Simultaneous Harmonics: 3

Stimulus Amplitude

 pp0[V]: 200m
 pp1[V]: 50m
 F0[Hz]: 25k
 F1[Hz]: 100k

General
 Start Analysis at Time[sec]: 150u
 Minimum Analysis Time At Each Frequency [sec]: 50u
 Stimulus Settling Time At Each Frequency [sec]: 50u
 Intermediate Node Name:
 Disable This Analyzer ☐

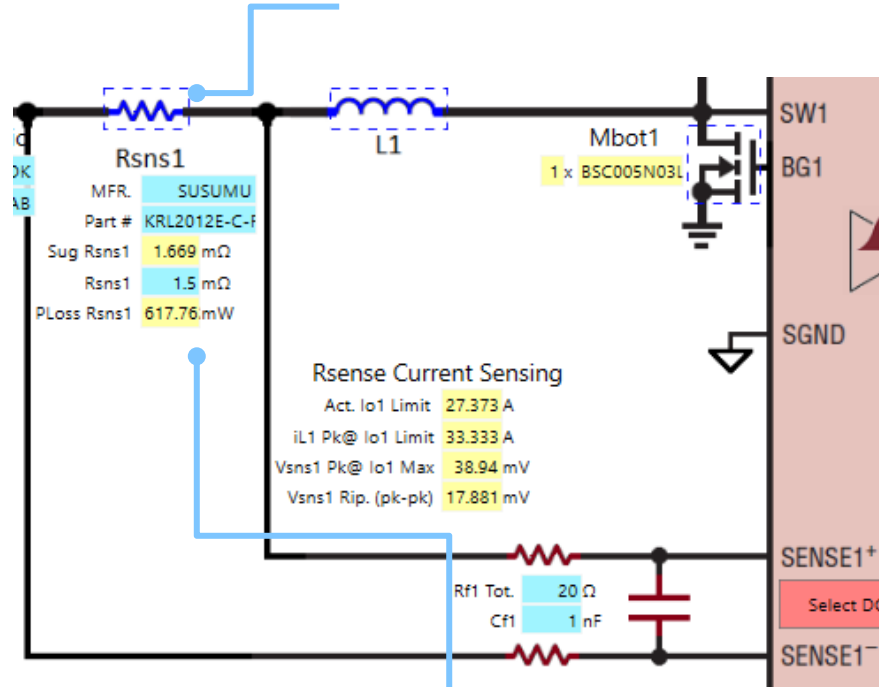
Calculated Duration (Simulated Time): 1.62712m sec

OK Cancel



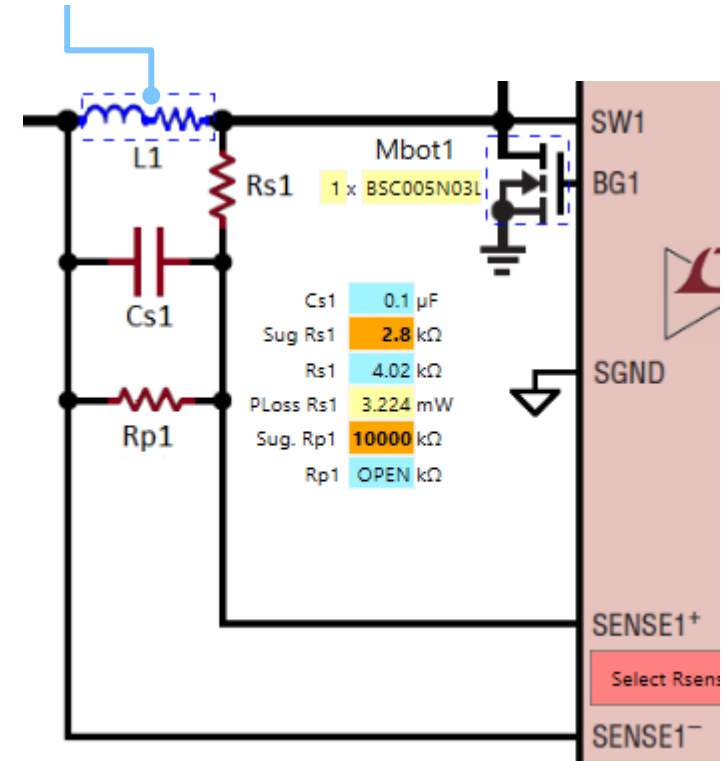
Buck Regulators: Current Sensing

- Rsense: Uses a sense resistor to capture inductor current



Remember to choose power rating around 2x this value

- DCR Sensing: Uses the DC resistance (DCR) of the inductor, set $R \cdot C = L / \text{DCR}$



Buck Regulators

Buck Components

- Current Sensing
 - Tradeoffs

R_{sense}

Pros

- Very accurate (1%)
- Low Tempco
- (100ppm/°C)
- Accurate even when
- inductor saturates

Cons

- Cost
- Heat dissipation

Note: R_{sense} cost may not be that high -
a 1W resistor might cost \$0.05

DCR Sensing

Pros

- Very accurate (1%)
- Low Tempco
- (100ppm/°C)
- Accurate even when
- inductor saturates

Cons

- Cost
- Heat dissipation

Note: A high-precision DCR (5%) may double the
cost of the inductor

Buck Regulators

Recommendations:

- Use Rsense whenever possible
Cost is similar (or even cheaper)
Relatively small size and efficiency penalty
- Only if the customer insists on DCR sensing
should you consider it
- Remember that you will need to increase the
current limit to take the Inductance
tolerance, DCR tolerance and Tempco into
account

Buck Regulators

Buck Switching

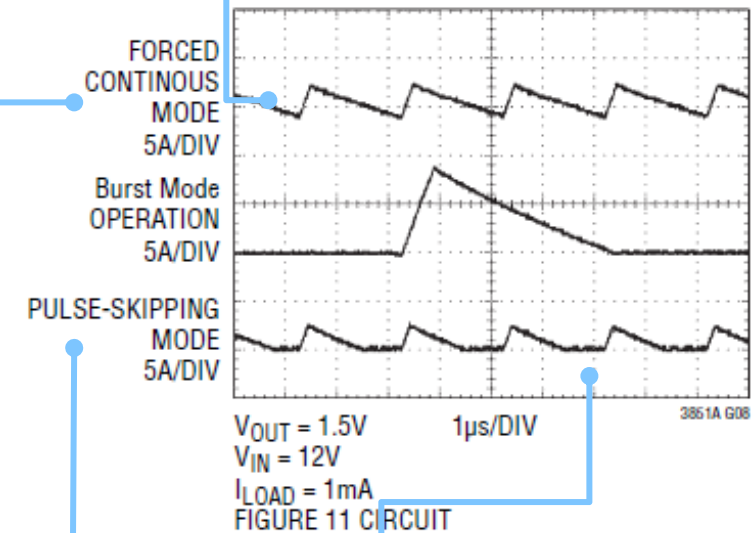
- Switching Modes:

- Forced-Continuous Mode (FCM)
 - Switching is the same at any load current. Inductor current waveform is the same, but the average current shifts up/down with I_{out}
 - Note that inductor current may become negative, and some converters have a negative current limit.
 - This mode has lowest V_{out} ripple, worst efficiency at light loads

- Pulse-Skip Mode
 - At lighter loads, the inductor current may hit zero. Controller will skip one or more pulses until V_{out} drops below some threshold, where switching resumes
 - Better light load efficiency, but slightly higher V_{out} ripple
 - F_{sw} is variable due to skipped pulses

I_L goes negative

Inductor Current at Light Load

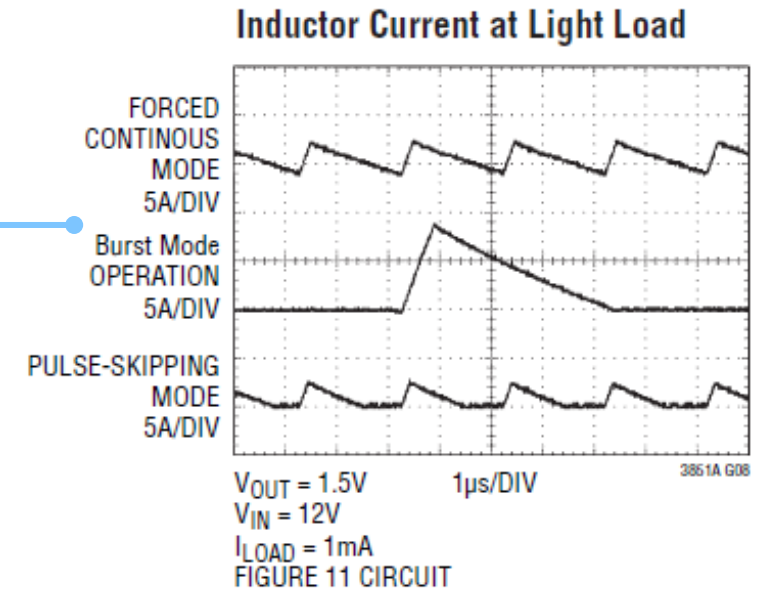


I_L stopped at 0A

Buck Regulators

Buck Switching

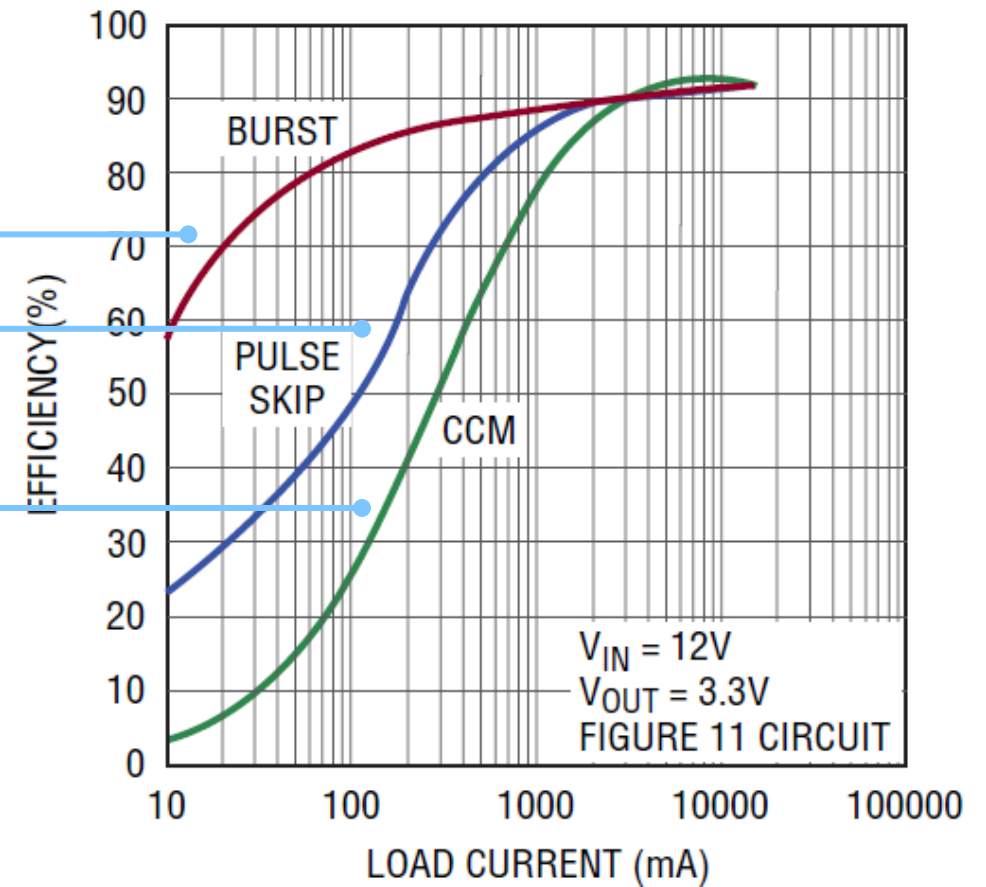
- Switching Modes:
 - Burst Mode
 - When load is light, and inductor current drops to zero, all parts of the controller are shut down, except for Voltage reference and output comparator.
 - When V_{out} drops below a pre-set threshold, switching resumes. This mode has the best light load efficiency, at the expense of higher V_{out} ripple. Effective F_{sw} is variable



Buck Regulators

Buck Switching

- Switching Modes:
 - Burst Mode
 - Pulse-Skip Mode
 - Forced-Continuous Mode

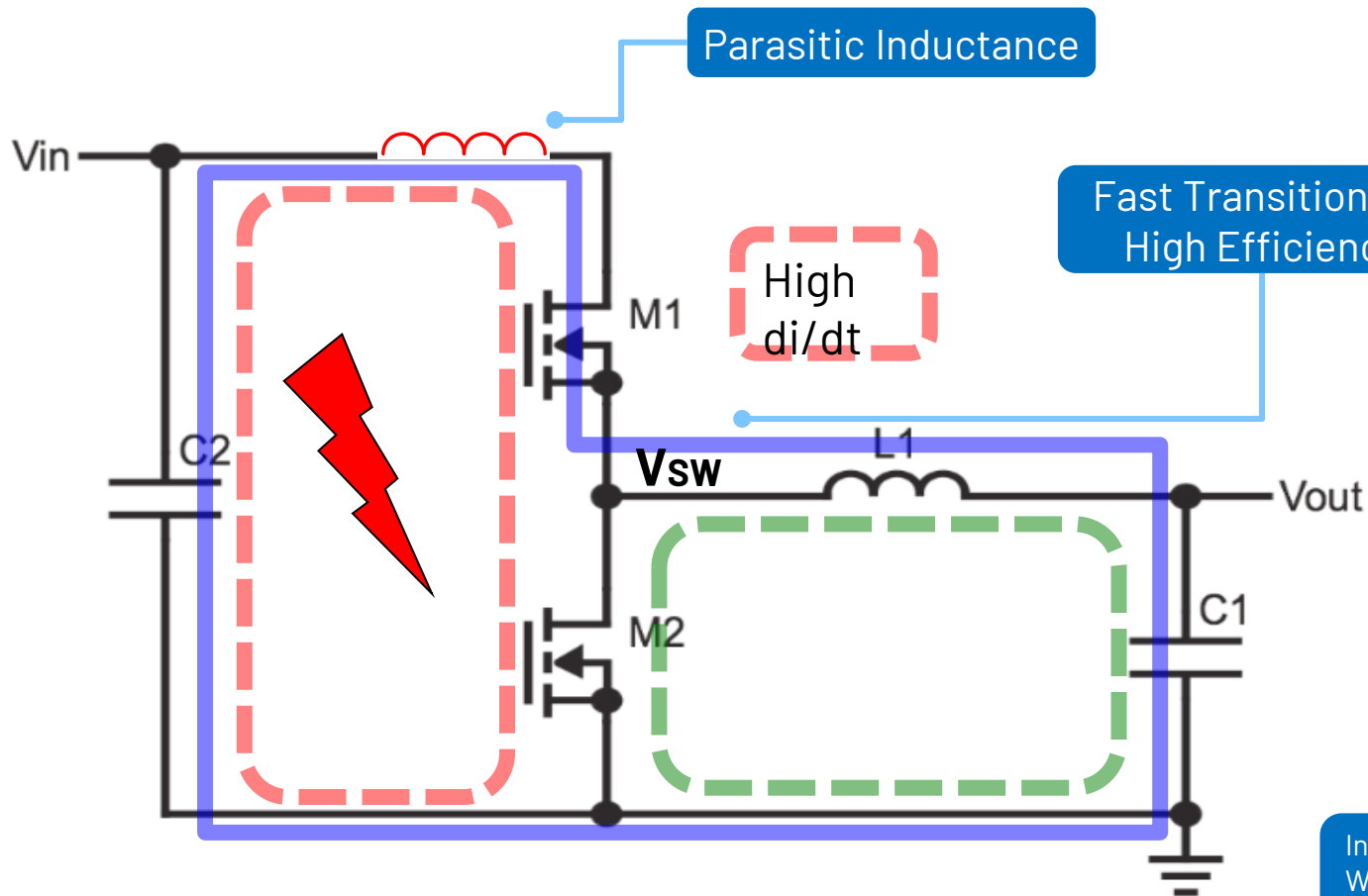


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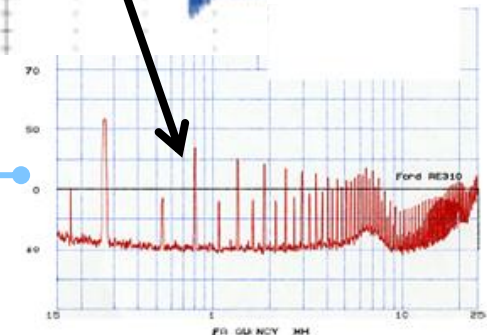
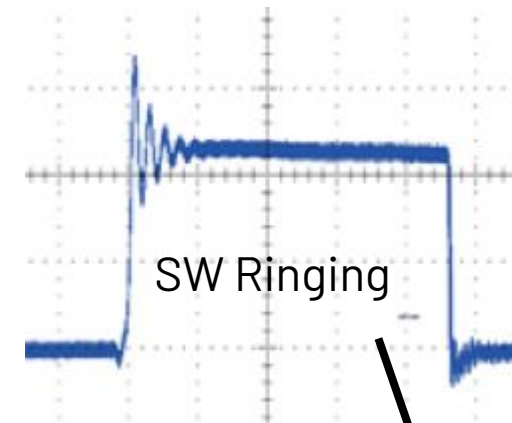
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Competitive Examples

- Silent Switcher™ switching regulators - ADI Patented technology
- Fast SW node transition needed for high efficiency



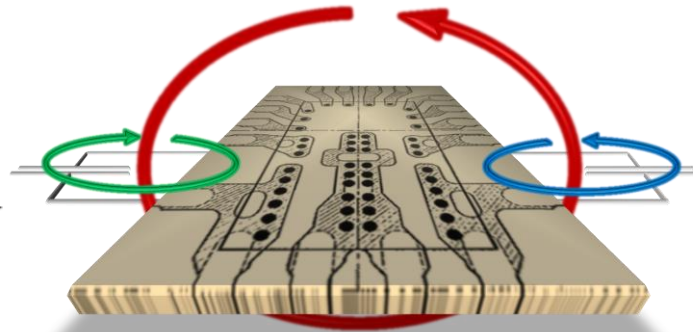
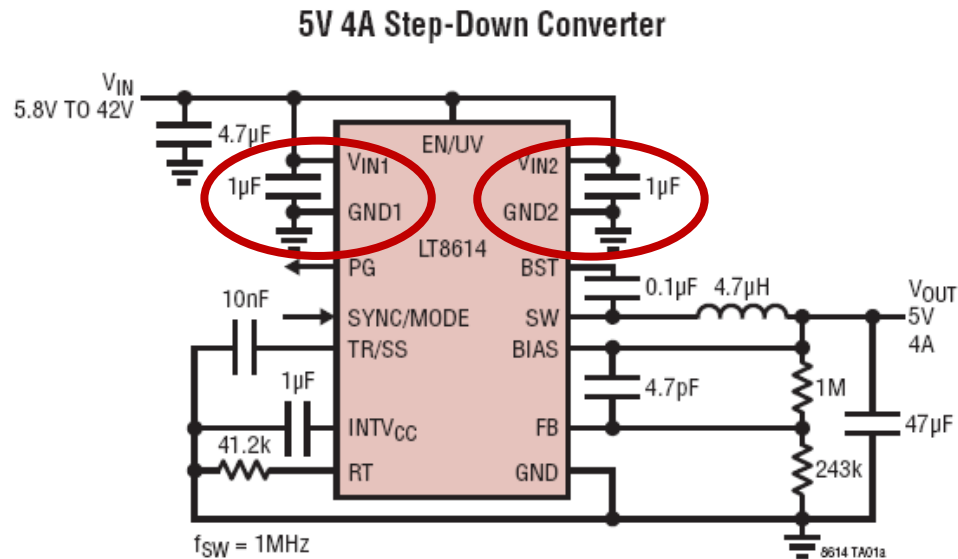
High di/dt and Parasitic Inductance in the Switcher Hot Loop Causes EM Noise and SW Ringing



In the Frequency Domain, We See High Frequency Harmonics

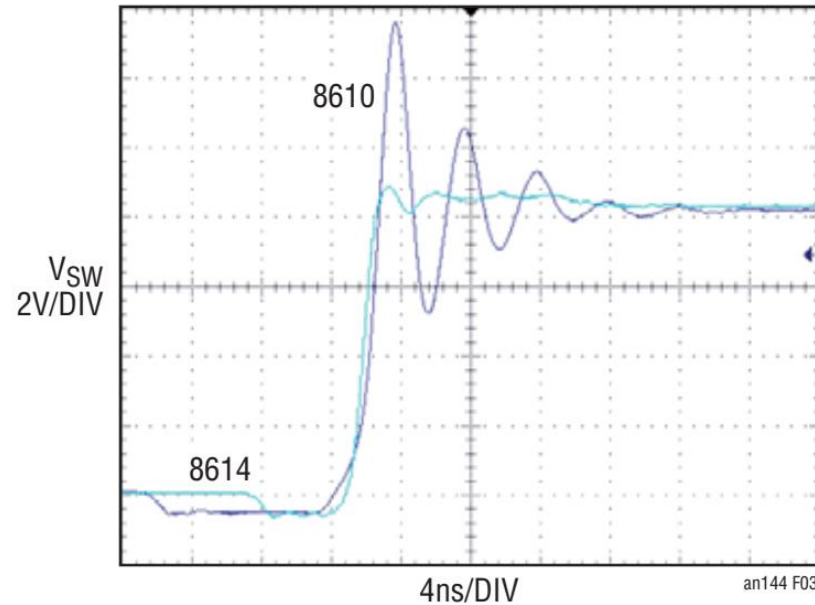
What's Behind Silent Switcher Technology?

- Reduce the magnitude of the harmonics by eliminating parasitics (**bond wire** → **flip-chip**)
- Reduce the energy in the harmonics by splitting the hot loops into **two smaller loops**
- Prevent the EMI from propagating by having the fields from the **two loops cancel** each other
- Internal switch drivers minimize switching power loss producing **fast, clean switching edges**

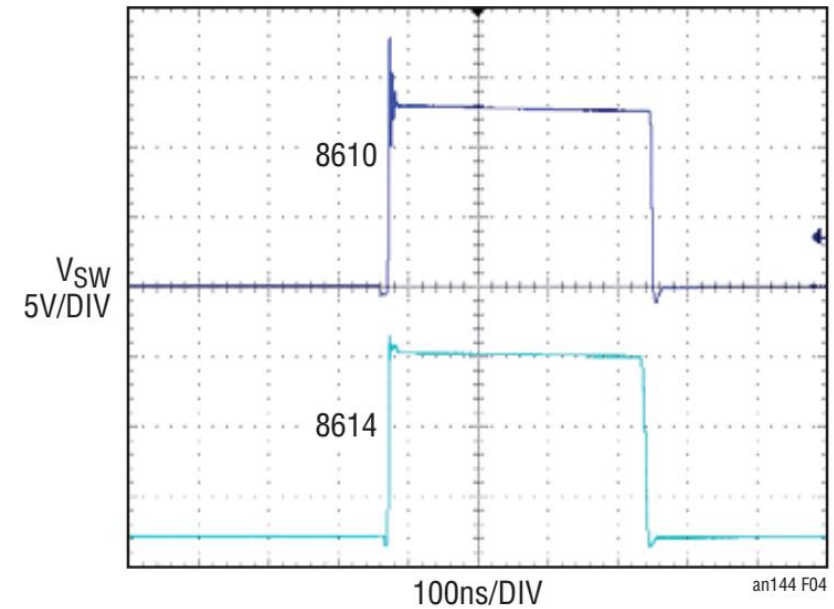


- **Silent Switcher® technology**
 - High efficiency
 - High switching frequency
 - Low EMI
 - All 3, **no trade-offs!**

Copper Pillar Flip-Chip and Magnetic Cancellation



**Figure 3. LT8610 and the LT8614, Switch Node Rising Edge
Both at $8.4V_{IN}$, $3.3V_{OUT}$ at 2.2A**



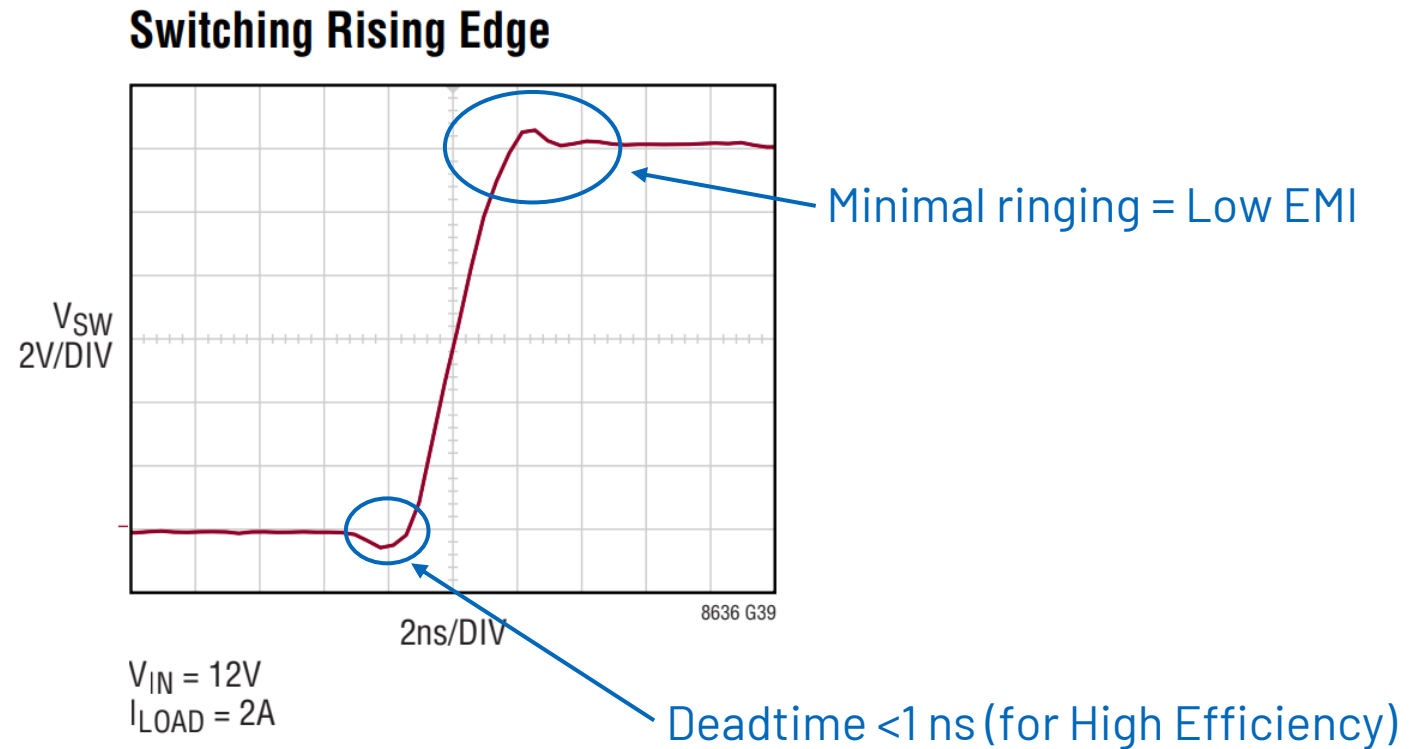
**Figure 4. LT8610 and the LT8614, Both at $13.2V_{IN}$,
 $3.3V$ 2.2A out**

LT8610: wire-bonded in MS16E

LT8614: Silent Switcher® 1 regulator: magnetic cancellation + copper pillar flip-chip

Rising Edge

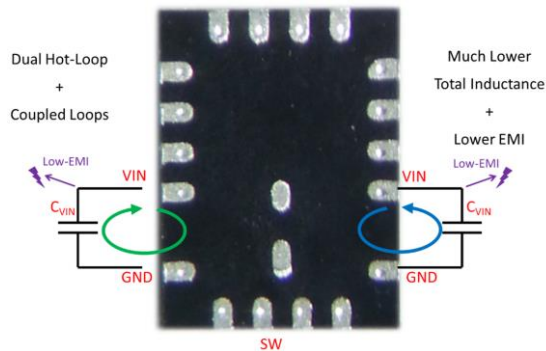
- Optimized powertrain for controlled slew rate and fast, clean switch edges



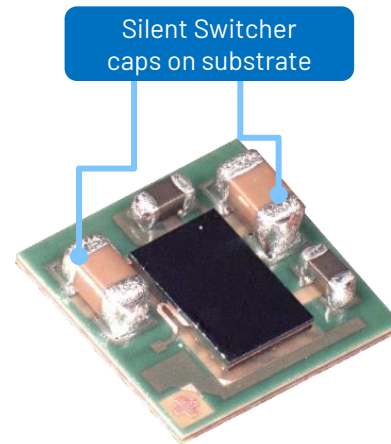
- SW edge looks great! Advertised in our data sheet!

New Silent Switcher 3 Technology

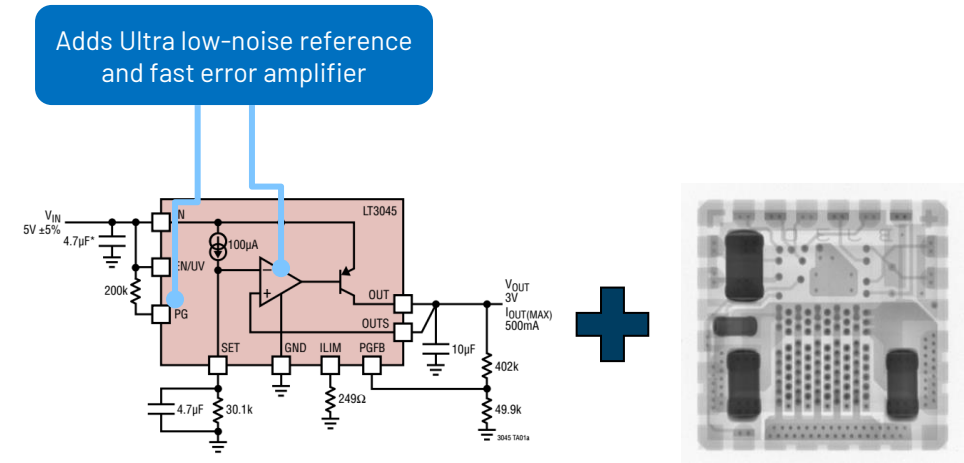
- Exceptional EMI performance
- High efficiency at high switching frequencies
- Easy PCB layout
- **Ultralow LF noise (10 Hz to 100 kHz)**
- **Ultrafast transient response**



Silent Switcher[®] 1 Technology

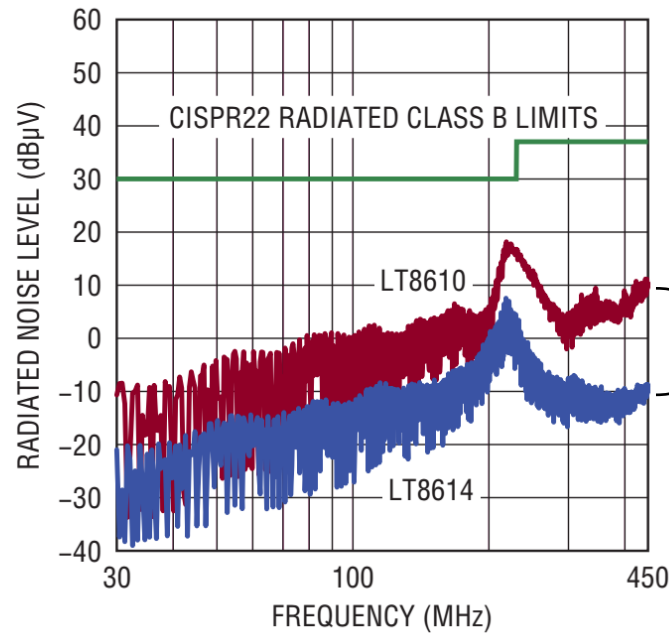


Silent Switcher[®] 2 Technology



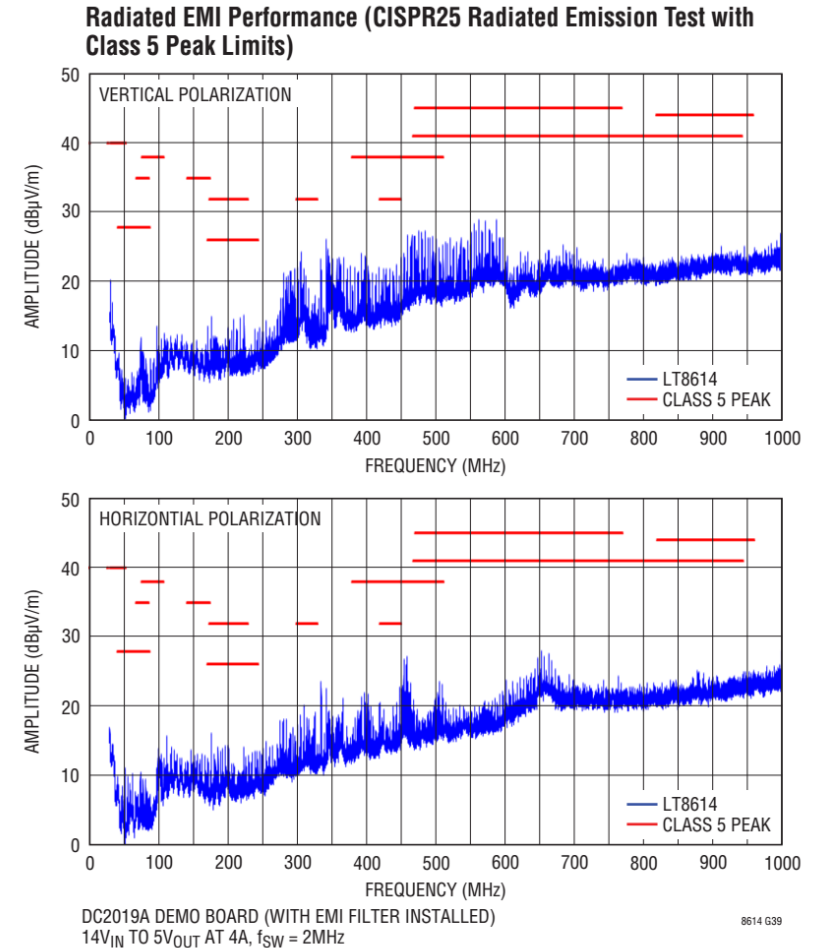
Silent Switcher[®] 3 Technology

Silent Switcher Radiated EMI Results



Silent Switcher[®]:
10 dB to 20 dB
improvement!

**Figure 2. LT8610 and LT8614 700kHz 14V to 3.3V 2A
Radiated EMI in GTEM Corrected for OATS**



LT8614 passes the most
stringent CISPR25 Class 5 limits

uModule

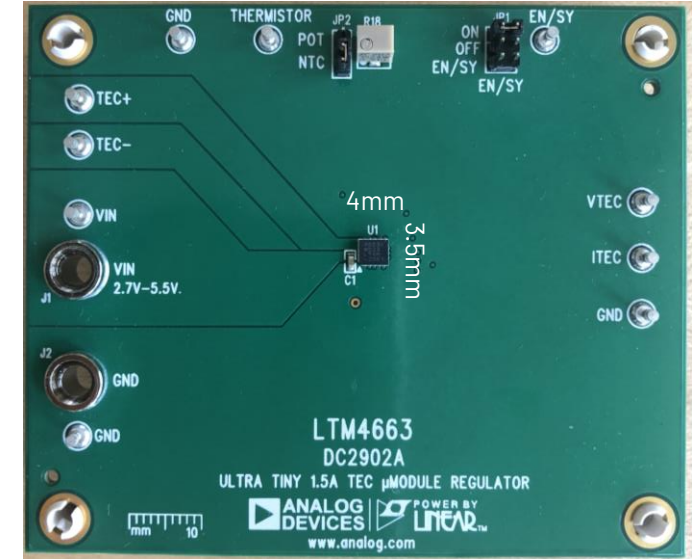


FEATURES

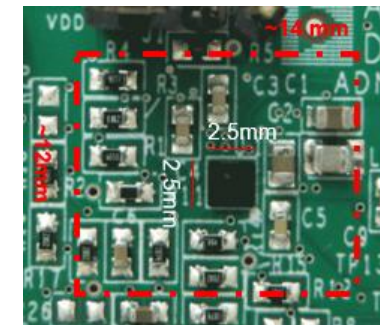
- Built-in Two Zero-Drift, Rail-to-Rail Chopper Amplifiers
- 2.7V to 5.5V Input Voltage Range
- 1.5A Driving Capability
- **1% Accuracy 2.5V Internal Reference Output**
- TEC Voltage and Current Monitoring
- Independent Programmable Heating and Cooling Current Limit
- Programmable Maximum TEC Voltage
- Default 2MHz Switching Frequency
- Synchronization from 1.85MHz to 3.25MHz
- Capable of NTC, PTC and RTD Thermal Sensors
- **3.5mm × 4mm × 1.3mm LGA Package**

APPLICATIONS

- TEC Temperature Control
- Optical Networking System, Optical Module
- LiDAR System



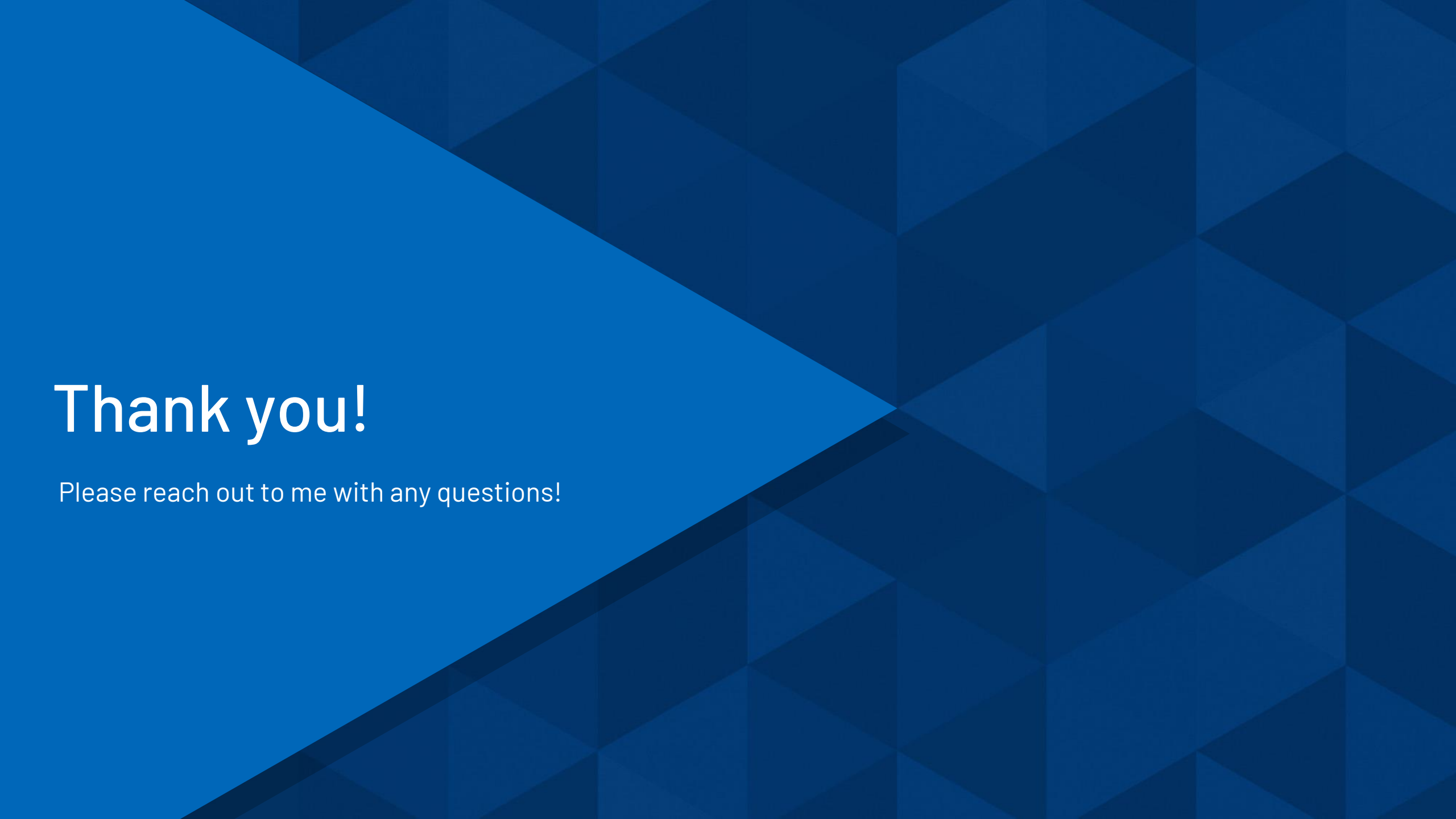
LTM4663



Discrete Solution

ADI eShop, re-imagined based on Customer Feedback

Features	Benefits
ADI + Legacy Maxim Products	• Singular experience for all ADI products and technologies • Single cart for products from ADI, including all Legacy Maxim products
Ship-Now or Scheduled Deliveries	• Ship-Now (within 1 week) from readily available inventory (higher price) • Or schedule your orders out to 2 years • Ship from inventory or ship to standard leadtimes
Scalable Pricing	• Pricing scales with quantity and/or immediacy of shipment • eShop allows higher volume discounted pricing. • Request for Quotes if your needs are >1K pieces
Contract/Negotiated Pricing	• Pre-existing negotiated pricing will extend to the eShop
Backlog Orders	• If no inventory, place backlog orders. • eShop will ship in standard leadtime or when part becomes available.
30-Day Terms or Credit Card	• Pay using credit cards (USD) • Or pay using Net 30-Day Terms with ADI directly • Auto \$10K Credit Limit Approval • Apply for 30-Day terms with ADI
Single order for 1pc or large volume	• Combine engineering prototype needs and volume purchasing in same transaction.
Full Order Management	• Manage orders including change delivery schedules, cancellations and check order status • View delivery dates, quote request status etc.
Human powered online chat	• Chat directly with actual people in Customer Operations • Human guided help across all regions – not an automated chatbot.

The background is a solid blue color on the left, transitioning into a pattern of dark blue and medium blue triangles and cubes on the right. A large, light blue triangle points towards the right, partially overlapping the geometric pattern.

Thank you!

Please reach out to me with any questions!

AHEAD OF WHAT'S POSSIBLE

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ADI *Power*